

Barriers to district heating deployment: insights from literature and experts

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ABSTRACT

Over the last century, district heating (DH) has only achieved global market shares of around 10 %. To reach its potential market shares of 25–50 % in Europe by 2050, growth must drastically accelerate. But deployment of DH is hindered by barriers, potentially slowing this fast-paced transition of the heat system. In this paper, we identify barriers to the deployment of DH through manual and GPT-aided literature reviews. These reviews are complemented by a survey of 94 DH experts from Europe and North America, regions which account for approximately 20 % of global- and 93 % of European DH supply. We find that economic and political barriers are considered both most significant and most difficult to overcome. The importance of individual barriers varies between countries based on the existing shares of DH and renewable heat in DH supply respectively. The most important individual barriers include high installation costs, regulatory uncertainty and insufficient policies to integrate DH with other energy sectors. If a faster pace of system-integrated DH deployment is desired, it is critical to reduce the underlying investment risk of district heating projects through an expansion of integrated energy system planning, and regulations such as zoning and mandates.

1. Introduction

Heating accounts for a third (nearly 145 EJ) of global energy demand (IEA, 2024a) and 80 % of CO₂ emissions in buildings globally (IEA, 2023), corresponding to 24 % of global energy-related greenhouse gas emissions (IEA, 2024a). The sector is therefore crucial for reasons of security of supply and decarbonisation. The IEA describes heating's current alignment with IEA's *Net Zero Emissions by 2050 Scenario* as "More efforts needed" (IEA, 2024b), while its sub-sector district heating (DH) is "Not on track" with "almost 4 % of global CO₂ emissions" (IEA, 2024c). This status aligns with the Draghi report's (2024) recommendation to "[...] extending acceleration measures and emergency regulation to heat networks [and] heat generators [...]". Consequently, the European Commission has scheduled publication of a Heating and Cooling Strategy by 2026 (European Commission, 2025).

Heating may be "less sexy" (Eggertson, 2004) than other energy sectors, and especially its lesser-known sibling DH merits a brief

introduction. In one definition, DH exploits the "use [of] local fuel or heat resources that would otherwise be wasted" (Frederiksen and Werner, 2013) such as industrial waste heat, waste-to-energy plants or combined heat and power generation, thereby increasing the energy efficiency compared to individual heating systems. Beyond the use of such sources, power-to-heat technologies (heat pumps and electric boilers) are increasingly becoming part of the mix (Grohnheit and Sneum, 2023). The main components of a DH system are the heat supply source and the pipe network which distributes the heat to residential-, industrial- and other types of end-users. Due to the additional cost for the network compared to individual heating systems, DH is primarily viable in areas with a sufficiently high heat demand density and a cheap and central source of heat supply.

DH is not by itself a low-carbon technology, as many different heat sources can feed into the DH network, a characteristic coined *X-to-heat* (Sneum et al., 2021). In fact, 90 % of global current heat for DH is based on fossil fuels (IEA, 2024c) and thus in need of large efforts if decarbonisation is to be achieved. Lund et al. (2014) have defined this

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Abbreviations:

CHP –	Combined heat and power
DH –	District Heating
GPT –	Generative pre-trained transformer
IEA –	International Energy Agency
PESTLE –	Policy, Economy, Social, Technology, Legal, Environmental
UK –	United Kingdom

pathway from high carbon to low carbon DH systems as a progression through generations of DH. Generations 1–3 were less energy efficient, at higher temperature and more carbon intensive than the low temperature, sector coupled and multi-source fourth generation.

DH supplied 9 % of global final heating needs in buildings and industry in 2022 (IEA, 2024c) and 13 % of Europe's heat demand in 2023 (Mathiesen et al., 2023). In Europe, studies have shown DH potentials of total heat supply in the range of 25 % (Billerbeck et al., 2024b) to 48 % (Mathiesen et al., 2023) by 2050. Whether the motivation for DH deployment derives from these potentials, getting on track with IEA's Net Zero Emissions pathway or the EU Renewable Energy Directive's ambition to "promote the development or modernization of efficient district heating and cooling networks" (European Parliament, 2023a), it is relevant to ask: what may hinder the deployment of decarbonised DH?

The study of barriers within DH is not new. Already in 1980, Jenkins (1980) found strong opposition against combined heat and power (CHP) from incumbent gas- and electricity industries. Recent years have seen more activity within the subject, such as Mazhar et al.'s (2018) state of the art of DH-study, which charted a large amount of challenges for DH. Similarly, the European Commission's (2022) landmark study, an overview of DH markets and regulations, and Abimbola (2022) provided an abundance of barriers to DH deployment. Barriers have specifically been addressed for the DH sector in the United Kingdom (UK) by Millar et al. (2019), the UK Department of Energy and Climate Change (2013) and for Germany by Köhler et al. (2024). Among the barrier studies covering specific themes within DH are Lygnerud's (2018) study of challenges in DH business models and Sneum's (2021) taxonomy of barriers to energy system flexibility in the DH sector. Sneum's study also provides the definition of "barrier" employed in our study: "a framework condition that inhibits a function, decision or behaviour" towards the desired result.

Our reviews in this paper identify a relatively high number of barriers distributed over a moderate number of documents (compared to e.g. Sneum's (2021) study of barriers to flexibility in DH with 100+ references). This indicates that barriers to DH deployment are well-described in scientific and grey literature. Despite this, until now they have not been systematically approached in the scientific literature. The reason appears to be that studies tend to focus on the solutions and potentials provided by DH but devote less attention to systematically addressing the barriers.

DH is inherently context specific, and its deployment is riddled with case-specific barriers that can never be fully mapped. The objective of this paper is thus not a comprehensive list of barriers, but a sufficiently wide and systematic understanding of barriers, to inform policymakers and the scientific community on respectively improving deployment conditions for DH and target further research towards areas needing attention. In doing so, the following research questions are addressed:

- What barriers to DH deployment can be identified?
- Are there tendencies or traits among the experts' findings on geography or technology, which merit a certain focus regarding the mitigation of barriers?

While we conclude with solutions to the most important barriers, we emphasise that the study is a review of barriers, not of solutions.

The paper is structured as follows: section 2 describes the methodology used. The results are presented in section 3 and discussed in section 4. Section 5 concludes with policy implications.

2. Methodology to identify and analyse barriers

We first define the concept of barriers before describing the process to identify barriers to DH deployment (cf. section 2.1). To balance the need for a comprehensive and objective identification of barriers, two main pillars constitute our study: A literature review (cf. section 2.2) and a comprehensive expert survey (cf. section 2.3).

2.1. PESTLE framework for the definition of barriers

While Boulding's (1978) first law dictates that "Anything that exists is possible", there is a mismatch between the actual deployment of DH and its technical potential, as exemplified in studies of Germany and Europe. In Germany, DH deployment is currently only seen in 70 % of high heat density areas (Manz et al., 2024), while Möller et al. (2019) identified sufficient heat sources available for DH to cover 59 % of Europe's 2015-heat demand. Gaps like these are well-known within energy research. Jaffe and Stavins, and DeCanio have described this as the energy efficiency gap (Jaffe and Stavins, 1994), or the energy efficiency paradox (DeCanio, 1998). Such gaps can be explained by barriers within the economic, institutional, psychological and social categories (Eyre, 1997; Sorrell et al., 2006).

We choose a similarly wide view on barriers. Examples of such wide frameworks include Ma et al.'s (2020) Smart DH Architecture Model, Sneum's (2021) taxonomy and the policy-oriented Political, Economic, Social, Technological, Legal and Environmental (PESTLE) framework. The latter was originally conceived for business studies by Aguilar (1967) as economic, technical, political, and social (ETPS) categories. Later adaptations have added the (L)egal and (E)nvironmental dimensions. PESTLE is in management literature defined as "A framework that categorizes and analyzes an important set of external forces [...] that might impinge upon a firm. These forces are embedded in the global environment and can create both opportunities and threats for the firm." (Rothaermel, 2013). We have chosen PESTLE, as it is well-established in management science and a proven framework for barrier analysis within fields related to the study, such as barriers for low carbon DH in the REWARDHeat project (Fransson et al., 2021), barriers to energy efficiency (Shilei and Yong, 2009) and decarbonisation investments in DH (Auvinen et al., 2023).

Beyond Sneum's (2021) definition of barriers ("a framework condition that inhibits a function, decision or behaviour"), four elements merit particular emphasis:

1. The distinction between actual barriers and perceived barriers. This matter is important but cannot be addressed in this review: validity-checking of barriers would require e.g. context-specific techno-economic analyses of individual cases.
2. The concept of "barriers" depends on the perspective. For policy-makers, the term "barriers" may be useful, while project developers and investors perceive the same barriers through the lens of "risk".
3. We know from Sneum and Sandberg (2018) that the impact of DH barriers is not uniform: some contexts may face a prohibitive barrier in e.g. taxation, whereas the same barrier might be a minor concern in other contexts. While we do suggest primary barriers (McAfee et al., 2004), we do so with this caveat in mind. For an applied impact assessment of barriers, a good example is in the Heat Roadmap Europe (Trier et al., 2018) juxtaposition of a barrier's presence against the potential for DH deployment in the given context.
4. Often, barriers will span several different PESTLE-categories. The categorisation of barriers is thus a normative choice by the authors,

based in a principle of interpreting the barriers regarding new “greenfield” and decarbonised DH deployment (while barriers may also apply to existing “brownfield” DH). For example, poor reputation of DH may result in both a lack of subscribers during the development phase, as well as customer defection during the operation phase.

2.2. Literature review of barriers to district heating

The literature review was performed using two approaches: first through a narrative literature review (cf. section 2.2.1), second with the assistance of a generative pre-trained transformer-tool (GPT) to screen a much larger corpus of literature (cf. section 2.2.2).

2.2.1. Narrative literature review

The narrative literature review identifies barriers within scientific and grey literature. The strength of narrative reviews is their exploratory approach that seeks to synthesize insights from a variety of perspectives and disciplines (Sovacool et al., 2018). This enabled reviews beyond the peer-reviewed literature that also include grey literature, e.g. European Commission reports, and thereby the sampling of barriers that were not captured in scientific studies.

A narrative review of a field as wide as “barriers to DH deployment” necessarily entails a delimitation, as the concept of “barriers” in its extreme extends into each point of friction within each interaction under each individual DH project. In contrast to stringent methods such as systematic reviews, the narrative review has no search protocol. The choice of barriers to include was thus the authors’ informed choice, a weakness in the method regarding potential bias (Sovacool et al., 2018). We mitigate this by contrasting the findings with those from two separate methods: the GPT-based review (section 2.2.2) and the expert survey (2.3). After the survey took place, the narrative review was supplemented with a structured review to strengthen the foundation in literature: querying Web of Science for journal articles published since 2020, using search terms “barrier” and “district heating”. This resulted in a gross list of 83, where a title-screening filtered this to 16 for further review.

To classify the barriers, we build on the PESTLE framework. As shown in Table 1, we extend this with sub-categories, including technologies affected (Technology), where the barrier is identified (Geography), at which decision-layer the barrier is identified (Level of origin), and the stage of the project at which it is encountered (Lifecycle level). These sub-categories are based in Sneum’s taxonomy (Sneum, 2021). Finally, the source is included to provide transparency on the validity. The full table is found in Appendix A.4.

2.2.2. GPT-based review

Acknowledging the potential strength of large language models, we have applied such a model in the review to expand the review beyond our own findings. The GPT-assisted review is based on GPT4All software (v. 3.2.1 using the Llama 3.1 8B Instruct 128k model). GPT4All became available for the study after the initial review and expert survey. This led to the iterations – manual review 1 – survey – manual review 2 – GPT-based review. GPT4All was chosen to additionally strengthen the review with DH-focused literature, beyond what was manually achievable. Having collected 2011 documents – peer reviewed articles and grey literature – related to DH in our own archive, we trained our model to query this highly specialized literature. This document set is unique in the sense that it is neither fully available in databases of peer-reviewed literature, since the set contains grey literature, nor in “web-scraping” GPTs, as these do not have access behind paywalls to the whole peer-reviewed body of literature. Since the software runs on a local computer, this approach also addresses the challenge of uploading potentially copyrighted material to a third party. The GPT4All-based review was conducted in August 2024. To provide clarity in the results, we split the results into our own and the GPT tool’s. While the manual review

provided the question framework for the survey, the GPT-based review was conducted to uncover barriers that would have escaped the authors’ manual review. Appendix A.5 shows the prompts and the corresponding answers from the GPT.

An important caveat with GPT tools is the risk of “hallucinations”, which are the capability of GPTs to provide seemingly correct but false responses. We addressed this in two steps, both described in detail in the results section (3.2.1). First, we screened for obviously incorrect statements. Second, we sampled the literature referenced by the GPT, to confirm the results. While both steps revealed shortcomings in the GPTs outputs, the GPT overall increased the number of identified barriers.

2.3. Expert survey

To complement the literature review, a survey was conducted in the second half of 2023 to collect information from DH experts. The experts were asked ten (open and closed) questions structured around the PESTLE framework and based on the results from the literature review. The open question was related to barriers to DH deployment in their own countries. Moreover, participants were asked in a closed question to rate barriers in terms of importance and difficulty using a five-point Likert scale, which is a psychometric scale commonly used in research that employs questionnaires (see e.g. Billerbeck et al. (2024a)). Importance refers to the significance or importance of a barrier for the future deployment of DH, while difficulty addresses how difficult it is to address or overcome a barrier. Table A1 in the Appendix shows the exact survey questions.

The survey was distributed by e-mail to the direct contacts of the author team, whereby about 80 experts were contacted directly. In addition, several associations and authorities were contacted, such as the International District Energy Association, Euroheat and Power, the European Heat Pump Association, as well as national associations in the different countries. These associations distributed the survey to their members, mainly through their newsletters. Finally, the online portal LinkedIn was used to circulate the survey. Due to the anonymous nature of the survey, it is not possible to quantify how many of the direct contacts responded to the survey or how many potential participants were reached through the newsletters or via LinkedIn.

In total, 94 experts from 20 European countries¹ and North America responded to the survey. This country coverage corresponds to approximately 20 % of global DH consumption (Werner, 2017) and up to 93 % of European DH consumption (European Commission. Directorate General for Energy. et al., 2022). Most of the responses to the survey come from Germany, Denmark, and Switzerland. In addition, most respondents work in academia, for a public authority or for a DH company. Table A2 in the Appendix gives an overview of the respondents per country and their main activity. The findings and conclusions come with the caveat of representing the above-mentioned geographies and experts. It is likely that future studies with representation of other geographies (e.g. China and Russia), and experts (e.g. investors) may reveal other responses.

The sample is analysed using descriptive statistics and statistical testing. For each Likert-scale question, the distribution of responses and their mean m are assessed for overarching trends regarding the importance and difficulty of individual barriers. Furthermore, a Mann-Whitney-U-test is applied to identify statistically significant differences between two independent sets of countries when the data are not necessarily normally distributed. All tests were conducted using the “stats” package in the R programming language.

The Mann-Whitney-U-test is based on rank sums, which are formed for the two (country) groups (see also Billerbeck et al. (2024a)). To

¹ Austria, Croatia, Czech Republic, Denmark, Estonia, Finland, France, Germany, Hungary, Ireland, Italy, Lithuania, Luxembourg, Netherlands, Norway, Poland, Slovakia, Sweden, Switzerland and the United Kingdom.

Table 1

The classification framework used in the review. Examples of sub-categories follow: Technology – Power to heat, Geography – Germany, Level of origin – Local, Lifecycle level – Development phase, and Source – Peer reviewed.

PESTLE category	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle level	Source
Political							
Economic							
Social							
Technical							
Legal							
Environmental							

calculate the Mann-Whitney-U-value, the rank sum R_1 of the first group with n_1 observations is used as shown in equation (1). To assess the relevance of a result, effect sizes (Pearson correlation coefficient r) and p-values with significance threshold of 0.05 are calculated. We use the z-score and the sample size n to calculate the correlation coefficient (cf. equation (2)). Thereby, the z-score is the z-standardised U value, calculated from the mean of the distribution μ_U and the standard deviation σ_U (cf. equation (3)).

$$U = R_1 - \frac{n_1(n_1 + 1)}{2} \quad (1)$$

$$r = \frac{|z|}{\sqrt{n}} \quad (2)$$

$$z = \frac{U - \mu_U}{\sigma_U} = \frac{U - \frac{n_1 \cdot n_2}{2}}{\sqrt{\frac{n_1 \cdot n_2 \cdot (n_1 + n_2 + 1)}{12}}} \quad (3)$$

3. Results: barriers from literature and the survey

We first present the barriers identified in the literature in section 3.1, then explore the findings of the survey in section 3.2, focusing also on differences between countries and technologies.

3.1. Barriers to district heating deployment based on the literature

The literature review provides a deep understanding of the barriers, as the literature typically describes barriers in further detail than the detail attainable through the survey. Section 3.1.1 presents the results from the narrative literature review, while section 3.1.2 focuses on the results from the GPT-based review.

3.1.1. Results from the narrative literature review

The manual review identifies 173 barriers, divided into 46 sub-categories from 58 sources under the main PESTLE categories. An expanded table of all barriers is found in Appendix A.4. This contains the technologies affected by the barrier, geographic context, level of origin and where they occur in the project lifecycle. For descriptions of individual barriers, we refer to the original references and Appendix A.4. Table 2 shows the six main (PESTLE) and 46 sub-categories, and references for each sub-category. We summarise these in the following. In the Discussion, we expand on the primary barriers identified through the survey, and how they relate to existing literature.

The political category concerns the lacking political framework for deploying DH. This includes the limited prioritization, e.g. in the absence of the long-term policies that are necessary for such capital-intensive infrastructure projects as DH. DH not being a focus of political strategies or subject to long-term decisions are mentioned and highlighted in several publications. Another aspect of the political barriers is poorly designed policy, exemplified by complicated financial support schemes.

Within the economic category, costs play a major role – both in terms of raising capital and operating the DH. A cost-related subject which requires a bit of nuance, is the distinction between high technology cost and lacking investment capital. While this may appear two sides of the

same coin, it is important to note that technology cost is a matter of innovation or bottlenecks, while capital can be challenging to obtain on the investor-side, e.g. in already-indebted municipalities, regardless of the technology cost. Furthermore, several issues regarding business models were identified, ranging from lacking profitability to needs for revised business models for new developments such as low-temperature and flexible consumers.

The social category is broadly divided into challenges with managing multiple stakeholders, and lacking acceptance and awareness of DH. This pertains both to end-users but also the incumbent energy suppliers and decision makers. This awareness, or *bounded rationality*, ties closely into the political barriers regarding absence of focus on DH.

The technical category can be defined into phases of deployment of DH: In the development phase, competition to other solutions is key, ranging from decreased heat demand through energy efficiency to alternative sources of heat supply. Choice of heat supply technology may be affected by path dependencies, e.g. by preferences for legacy combustion-based solutions. But even before this stage, DH is challenged by its distinct difference from e.g. electricity and gas: while the other networked energy sectors can draw on a national or international set of demand and supply, and typically utilises already-existing network infrastructure, DH must establish both supply and customers from a relatively smaller local base, and then construct the whole network infrastructure from scratch. As technologies related to DH are mostly mature, the barriers relate less to the supply-side, but more to the demand. Here, the lacking short-term ability to manage demand, e.g. through IT-solutions, and the long-term uncertainty of demand-deflection are central.

The legal category ranges from regulatory uncertainty, to the difficulty of lifting the administrative burden on both the DH and regulator-sides. Several studies noted the lack of regulatory standardisation, ranging from environmental issues arising from improper fuel categorisation, to contracting mechanisms, and the absence of lifecycle costing in cost-benefit comparisons between heating investments.

Finally, the environmental category concerns the constraints induced by the surrounding environment. For example, the lack of physical space for thermal storage, the limitations in resource availability brought upon by uncertain or variable energy sources or the low temperature of excess heat, requiring transition from third to fourth generation DH (see Lund et al. (2014) for description of DH generations).

Out of 46 sub-categories, five were only identified in grey literature. When they are excluded, a strong focus on economic (7) and technical (13) barriers is identified in the peer-reviewed literature. In contrast, the political, social and environmental dimensions combined only cover 16 sub-categories in the published literature.

3.1.2. Results from the GPT-based review

Overall, the GPT-based review identified 36 barrier categories, whereof 16 were additional to the initial barriers identified in the manual review (Table 3). The GPT-based review thus substantiates our existing review-findings and provide new ones that are presented in the following.

A few of the GPT categories stuck out (* in table), meriting a closer inspection (see Appendix A.5 for additional information on these).

Table 2

The main PESTLE-categories and the sub-categories that barriers to DH deployment have been organised into. Sub-categories marked with * note those that were identified in a subsequent review iteration, after the survey taking place. Barriers solely identified in grey literature are marked with **. The survey category “High air pollutant emissions” was subsequently changed to “Negative environmental externalities” to encompass a wider array of barriers.

Reference numbers in the table refer to the following sources: 1 (Chassein and Roser, 2017), 2 (Department of Energy and Climate Change, 2013), 3 (European Commission. Directorate General for Energy. et al., 2023), 4 (Andrews et al., 2012), 5 (Bolton and Foxon, 2015), 6 (Salling, 2024), 7 (Bauknecht et al., 2017), 8 (de la Vega and Mišech, 2022), 9 (Lowes et al., 2018a), 10 (Lowes and Woodman, 2020), 11 (Millar et al., 2019), 12 (Stærke, 2022), 13 (Gullev and Duedahl, 2019), 14 (Ritzau, 2022), 15 (Mazhar et al., 2018), 16 (Sneum, 2020), 17 (Danish District Heating Association, 2023), 18 (Thomaßen et al., 2021), 19 (Gjoka et al., 2023), 20 (BREUniversity of EdinburghCentre for Sustainable Energy, 2013), 21 (Reintoft, 2023), 22 (Trier et al., 2018), 23 (Lygnerud, 2018), 24 (European Commission. Directorate General for Energy. et al., 2022), 25 (Fransson et al., 2021), 26 (Zaunbrecher et al., 2016), 27 (Sneum, 2020), 28 (Jenkins, 1980), 29 (Lowes et al., 2018b), 30 (Sneum, 2021), 31 (Dixon and Gielen, 2006), 32 (Heat Pump Centre, 2019), 33 (Werling, 2018), 34 (Viuff, 2024), 35 (Stöckel and Sneum, 2023), 36 (IRENA, 2017), 37 (Lake et al., 2017), 38 (Bull and Eadson, 2023), 39 (Herreras Martínez et al., 2022), 40 (Nie et al., 2023), 41 (Devenish and Lockwood, 2024), 42 (Arriazu-Ramos et al., 2021), 43 (Schmidt and Leitner, 2021), 44 (Marszał-Pomianowska et al., 2024), 45 (Frölke et al., 2022), 46 (Gürsan et al., 2024), 47 (Herreras Martínez et al., 2025), 48 (Meirbekova et al., 2024), 49 (Lygnerud et al., 2022), 50 (Korucan et al., 2024), 51 (Pakere et al., 2023), 52 (Guzzini et al., 2020), 53 (Reda et al., 2021), 54 (Fritz et al., 2022), 55 (Santin et al., 2020), 56 (Pakere et al., 2021), 57 (Jodeiri et al., 2022), 58 (Toffanin et al., 2021).

PESTLE	Sub-category	References
Political	Not a focus of political strategies	7,8,9,10,11,12
	Missing long-term decision making	4,5,6,41,46,47,48,49
	Lacking system integration policies	2,4,45
	Limited financial support is available	20,50
	Complicated financial support schemes	1, 44
Economic	High upfront capital expenditure	3,4,8,16,17,18,40,47,48, 50,51,52
	High finance costs	5,13
	High installation costs**	14
	High operational costs	11,15,16
	High end-consumer prices	22,35,39
	High costs for backup capacity**	3
	Lacking financial experience	8,19
	Capital lacking or unavailable*	11,20,21,22,42,47
	Outdated business model*	22,23,43,44,45,49,51,52,53
Social	Low acceptance of consumers	3,11,20,22,25,26,39,48, 50
	Low awareness of consumers of benefits	3,22,25,26,27,38,42,48,54
	Status quo bias	3,4,20,22,28,29
	Multi-stakeholder challenge and high variety of stakeholders	3,20,43,46,53,54
	Missing knowledge and skills gap	20,53
Technical	Low technical maturity	3,45,48,50,53,57
	Low market maturity	3,12,15,47,49
	Competition between supply technologies	2,3,8,22,25,39,41,42,55
	Cybersecurity*	43
	Lock-in effect and path dependency	11,25,43,51,53
	Low suitability of the building stock	3,24,42,52
	Low connection rates to DH grids**	35
	Data on energy sources constrained*	3,24,39,48,51,56

Table 2 (continued)

PESTLE	Sub-category	References
	Demand management insufficient*	11
	Demand uncertain*	3,20,23,25,40,43
	Demand transparency insufficient*	42
	DH requires new infrastructure, gas/electric sector does not*	4,52
	Long deployment phase*	15
	Source and sink not co-located*	3,15,55,56,57
Legal	Regulatory uncertainty	8,25,41,43,44,54
	Difficult permitting process and administrative burden	3,24,43,48,50,53
	Lacking standardization	15,19,20,22,49
	Lacking capacity of authorities	1,3,5,20,39
	Lacking ex-post accountability of energy decisions*	42
	Bans or mandates for district heating	24,30,52,58
	No third-party regulations	3,30,54
Environmental	Limited resources	3,8,15,20,39,43,53
	Limited land availability	3,16,33,39
	Negative environmental externalities	36,48
	Limited access to the power grid**	31,32
	Limited fuel availability	37
	Excavation problematic*, **	34

Reviewing the GPT's background literature (BREUniversity of EdinburghCentre for Sustainable Energy, 2013) on *social housing concerns* showed that the GPT misinterpreted it, by claiming an “impact of DH on fuel poverty, heat prices, or other issues related to energy affordability”. While the original text showed instead that it was a matter of prioritising focus on reducing fuel poverty, over installation of individual heat meters. Also, the GPT output includes a direct error in the response, “Traditional DH systems operate at high temperatures (around 80–90 °C), making it difficult to integrate renewable heat sources like solar thermal or biomass.”. It is typically neither problematic to reach 80–90 °C for DH provided from biomass (Danish Energy Agency, 2024) nor from solar (Abrecht et al., 2024). The GPT raising this as an issue may reveal a wider misconception regarding the ability of renewables to deliver sufficiently high temperatures.

Especially the technical category stands out in terms of additional barriers. Here, the DH generations (as defined by Lund et al. (2014)) are in strong focus, partially through the scope of infrastructure age and supply temperature. The legal category provides a relevant addition, as *Taxation policies* (unlevel taxation of heat technologies), *Social assistance programs* (social policy-based subsidies for certain heat types may create an unlevel playing-field) and *Price regulation* (price regulation inhibiting cost recovery) were not explicitly emphasised in the manual review. All refer to regulation that potentially directly inhibits DH deployment. Finally, the environmental category demonstrates a slightly different approach to the question than in the manual review. The findings there are essentially a list of potential pollution vectors (i.e. waste, noise, air).

3.2. Barriers to district heating deployment according to experts

This section presents the results of the survey. Section 3.2.1 discusses the barriers to further DH deployment using the PESTLE framework, before sections 3.2.2 and 3.2.3 further delve into the differences between countries and technologies.

3.2.1. Overview of perceived barriers

(i) Insights from open question

Table 3

Findings from the GPT-based review. Categories from the manual review (left – “Sub-category”) matched against the GPT-based review (right – “GPT category”). While the GPT model references the documents on which it derives results, sources are not allocated to individual pieces of information. The list of references thus applies to the answers in the table (Abimbola, 2022; BEIS, 2020; BREUniversity of EdinburghCentre for Sustainable Energy, 2013; Euroheat & Power, 2015; European Commission. Directorate General for Energy et al., 2022; Hanna et al., 2016; IRENA and REN21, 2020; Werner, 2006). All of these are grey literature. Barrier categories that we consider doubtful are marked with *. Where a GPT-category under one PESTLE-category aligned with a similar PESTLE-category in the manual review, the GPT-category was placed there, noting in parenthesis the GPT's original PESTLE-category. See Appendix A.5 for details.

PESTLE	Sub-category	GPT category
Political	Not a focus of political strategies	Lack of clear policy and regulation
	Limited financial support is available	Insufficient funding; Lack of investment incentives (Economic)
Economic	High upfront capital expenditure	High upfront costs
	High operational costs	Fluctuating fuel prices Low economies of scale High debt levels for households and businesses
Social	Low acceptance of consumers	Public perception and acceptance; Community engagement and buy-in
	Low awareness of consumers of benefits	Limited public awareness and understanding (Political); Lack of awareness and education
	Multi-stakeholder challenge and high variety of stakeholders	Resistance from existing stakeholders Social housing concerns*
Technical	Competition between supply technologies	Competing energy sources (Political)
	Low suitability of the building stock	High operating temperatures; Incompatibility with renewables* Ageing infrastructure Low efficiency Insufficient district heating pipe networks Inadequate heat storage facilities
Legal	Regulatory uncertainty	Lack of clear regulations
	Difficult permitting process and administrative burden	Planning and licensing issues
	Lacking standardization	Lack of standardization (Technical)
	Bans or mandates for district heating	Zoning laws and planning regulations (Political)
	No third-party regulations	Insufficient third-party access (TPA) rules Inconsistent pricing policies Taxation policies Social assistance programs Price regulation
Environmental	Limited resources	Resource extraction and processing
	Limited land availability	Land use and habitat disruption Water usage Waste management Air pollution Climate change mitigation Noise pollution

The first survey question was designed as an open question to elicit barriers without the explicit framing around PESTLE. The related question was “In your opinion, what are the main barriers to the further deployment of district heating in your country?”. In total, 83 respondents indicated barriers for their respective countries. Their

answers were analysed according to the PESTLE framework presented in Table 2, resulting in Fig. 1. The most frequently stated barrier is a lack of political support and long-term decision-making. Respondents have mentioned “[a]n incoherence among targets at different levels (EU, national, regional, local)” or a “lack of central commitment to the development of district heating networks”. Consequently, “District heating is not politically supported”, even though it is “clearly acknowledged in strategies”.

Beyond this, there is considerable heterogeneity between the barriers that were raised by the survey respondents. “[F]inancing challenges” resulting from “too high” rate of return expectations are a first example in this group. The negative perception of DH is another frequently raised aspect, explained by “monopoly and high prices” and a “preference for decentralized energy systems”. Further challenges to DH are identified from existing gas networks and from the emergence of heat pumps as a decentralized competitor. While regulatory uncertainty, especially concerning the price design, is also mentioned by a few respondents, legal and environmental barriers were raised less commonly in the initial open question. In addition, the literature has identified certain barriers that were not raised by our respondents, including higher operational costs, uncertain future demands, and low market maturity.

(ii) Insights from closed questions

Furthermore, the perception of different barriers was captured along the PESTLE framework with the closed multiple choice question “How important/difficult do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from (1) not important to (5) very important.”. As Fig. 2 (A) demonstrates, economic barriers were perceived as the most important, followed by political and environmental barriers. A similar pattern can be observed regarding the barrier's difficulty in Fig. 2(B), whereby economic and political barriers were followed by legal barriers.

Fig. 3 compares the two dimensions in a matrix, contrasting the importance on the y-axis with the difficulty on the x-axis. This figure makes it evident that economic and political barriers were perceived as particularly important and difficult at the same time.

In addition to the overall rating, participants were asked to rate specific barriers per PESTLE category in terms of importance and difficulty (see Table 2 for the list of barriers). The results of this rating are presented in Appendix A.3. In the category of policy barriers, respondents rated the “lack of system integration policies” as the most important, followed by “missing long-term decision-making”. At the same time, these two barriers were rated as the most difficult to overcome. Compared to all barriers listed, the two economic barriers “high upfront capital expenditure” and “high installation costs” (e.g. labour, certification) were rated as the most important barriers. “High installation costs” were also perceived as the most difficult to overcome. This result is in line with the overall rating of the PESTLE categories. The most relevant social barrier seems to be the “multi-stakeholder challenge and high variety of stakeholders” involved in DH. Of all listed social barriers, it was rated as most important and most difficult to overcome. Low awareness of consumers of benefits ranked second in terms of importance. The technical barrier with the highest rating of importance and difficulty was the “lock-in effect and path dependency”. “Low connection rates to DH” ranked second highest in the technical barriers category. From a legal perspective, “regulatory uncertainty” was seen as the most important and difficult barrier. Of all the barriers listed, it ranks third, after the two economic barriers. Thus, in addition to economic viability, a stable and secure regulatory framework is very important for successful DH markets. Another legal barrier with a higher rating in terms of importance is “difficult permitting process and administrative burden”. Additionally, “lack of capacity of public authorities” was also rated rather high in terms of difficulty. Finally, in the category of environmental barriers, respondents rated “limited land availability” as most important and difficult to overcome. Compared to

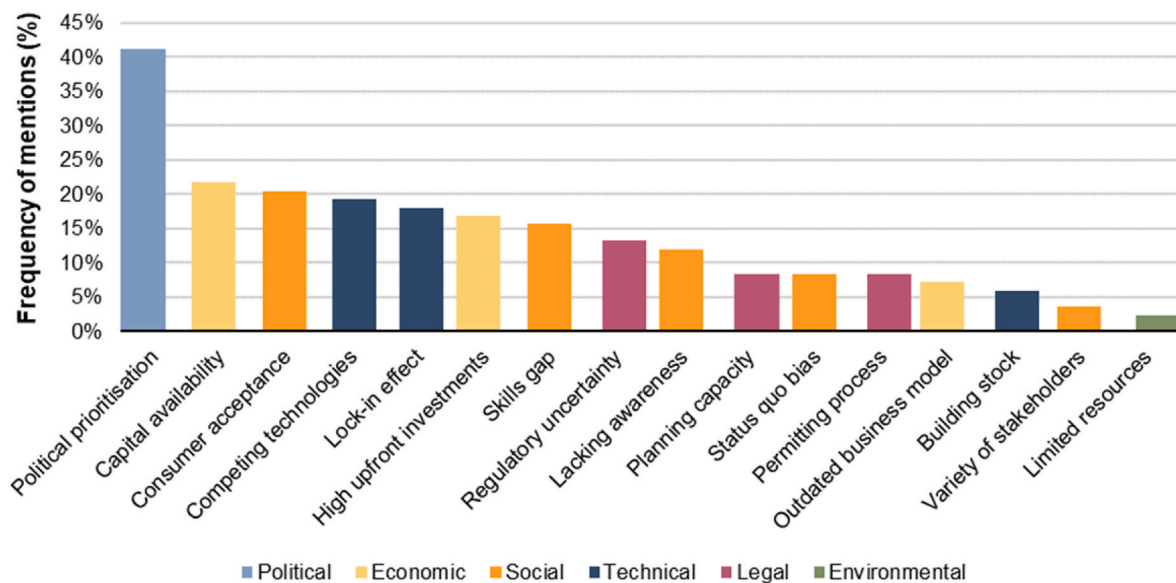


Fig. 1. Share of respondents identifying each of the barrier for the further DH deployment. The responses are categorized according to the framework presented in Table 2. Survey question: In your opinion, what are the main barriers to the further deployment of district heating in your country?.

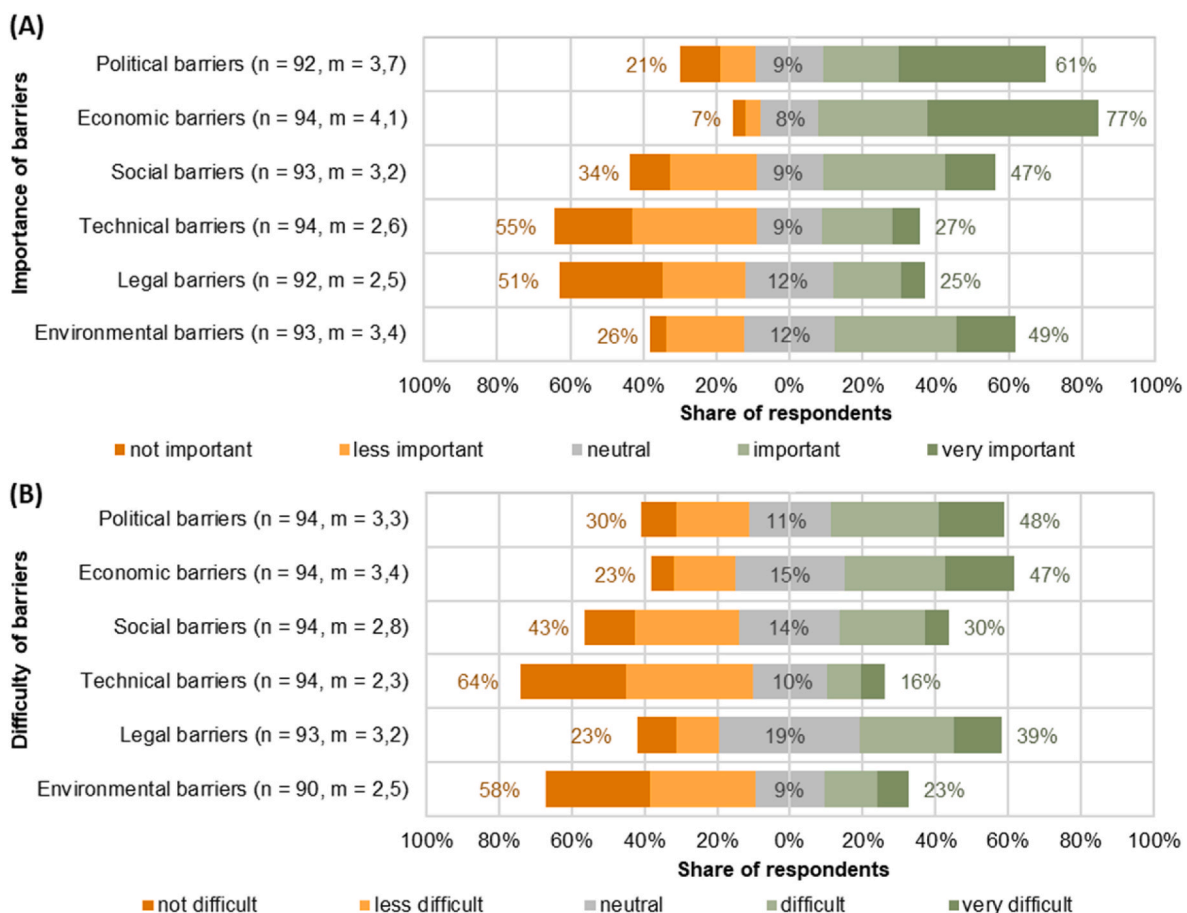


Fig. 2. Overview of perception of barriers for the further DH deployment. (A) Showing the rating of importance and (B) showing the rating of difficulty. n = number of respondents, m = mean, numbers on the left and right show sum of positive and negative ratings. Survey question: How important/difficult do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from (1) not important to (5) very important.

all barriers, environmental barriers were perceived as less important but rather challenging.

3.2.2. International differences

There are some clear differences between the countries regarding the perceived barriers. For example, social barriers are rated as more

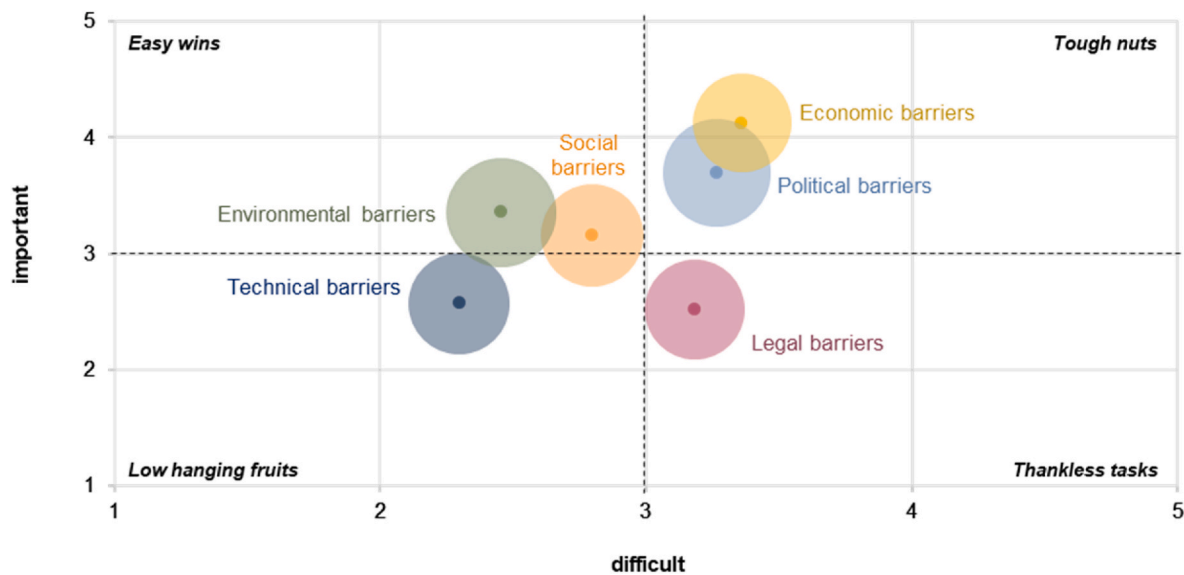


Fig. 3. Matrix of perception of barriers for the further DH deployment. Size of the circles shows the standard deviation. Dotted line shows medium/neutral rating. Survey question: How important/difficult do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from (1) not important to (5) very important.

important in Germany than in other countries, while competition from decentral heating solutions (i.e. electric boilers and heat pumps) is rated as more important in Norway and Sweden compared to other countries.

As there are only a few responses for several countries, the countries were divided into groups and the differences between the groups were systematically analysed using test statistics (cf. Fig. 4 and test statistic in

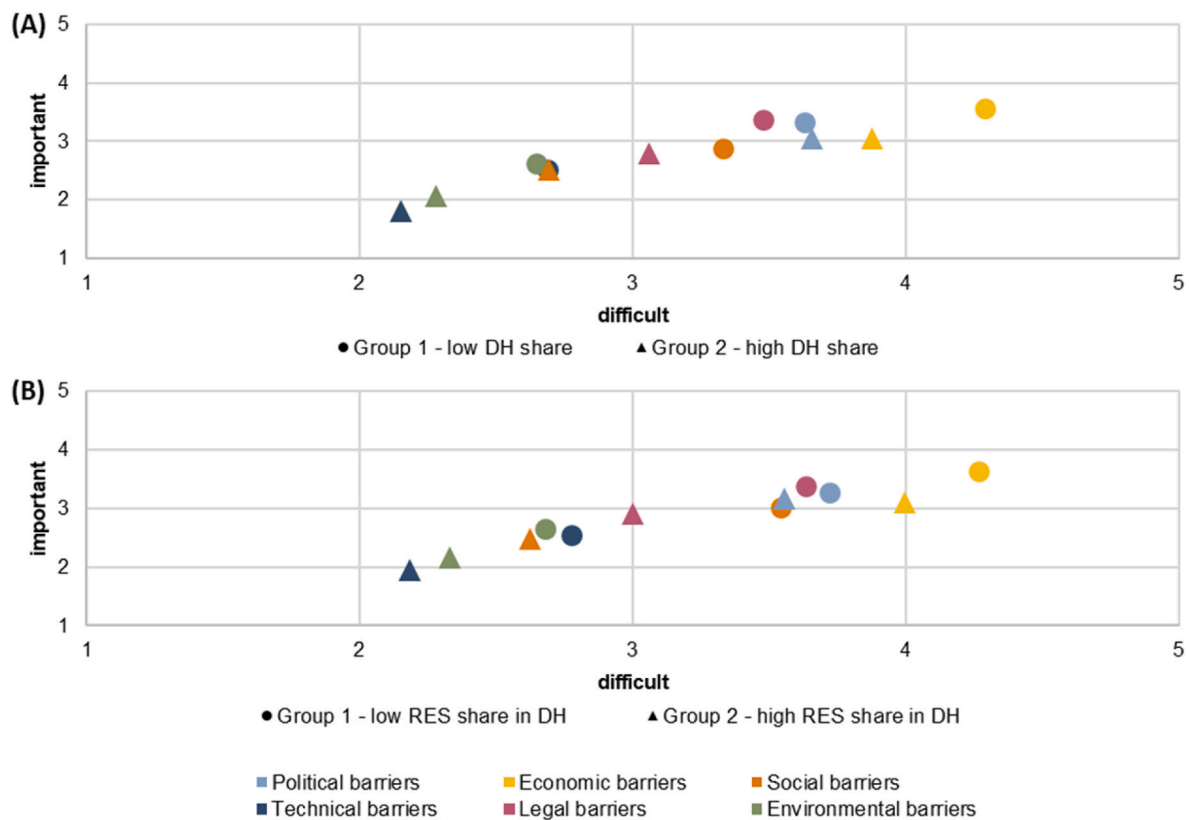


Fig. 4. Overview of perceived barriers for the further deployment of DH by different country groups. (A) Country groups based on share of DH with group 1: Germany, Switzerland, Netherlands, Austria, Hungary, Ireland, United Kingdom, France, Norway, Croatia, Italy, Luxembourg and group 2: Denmark, Finland, Sweden, Estonia, Lithuania, Poland, Czech Republic, Slovakia. (B) Country groups based on share of renewables in DH with group 1: Germany, Hungary, Ireland, United Kingdom, Norway, Slovakia, Croatia, Italy, Luxembourg, Poland, Netherlands, Czech Republic and group 2: Denmark, Finland, Sweden, Estonia, Lithuania, Austria, France, Switzerland. Survey question: How important/difficult do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from (1) not important to (5) very important.

Table 4

Test statistics of group comparisons. p-value significant level <0.001*** <0.01**; <0.05*; >0.05; r effect size 0.2 small; 0.5 medium; 0.8 large; significant effects are bold. Survey question: How important/difficult do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from (1) not important to (5) very important.

Variables			Mann-Whitney-U-value	p-value	Effect size r
Group 1 - low share of DH vs. Group 2 - high share of DH	Importance	Political barriers	901	0.954	–
		Economic barriers	1059	0.162	–
		Social barriers	1177	0.017*	0.26
		Technical barriers	1140	0.039*	0.22
		Legal barriers	1107	0.077	–
		Environmental barriers	1041	0.241	–
	Difficulty	Political barriers	1014	0.349	–
		Economic barriers	1118	0.063	–
		Social barriers	1066	0.160	–
		Technical barriers	1222	0.005**	0.30
		Legal barriers	1127	0.049*	0.21
		Environmental barriers	1142	0.038*	0.22
Group 1 - low share of renewables in DH vs. Group 2 - high share of renewables in DH	Importance	Political barriers	1072	0.370	–
		Economic barriers	1064	0.391	–
		Social barriers	1382	0.000***	0.38
		Technical barriers	1261	0.012*	0.27
		Legal barriers	1285	0.006**	0.29
		Environmental barriers	1113	0.215	–
	Difficulty	Political barriers	1003	0.764	–
		Economic barriers	1197	0.049*	0.21
		Social barriers	1226	0.026*	0.24
		Technical barriers	1248	0.015*	0.26
		Legal barriers	1180	0.065	–
		Environmental barriers	1151	0.116	–

Table 4, data for country groups based on (European Commission. Directorate General for Energy et al., 2022)).

The first group comparison uses the share of DH in the countries to divide the countries into two groups (cf. Fig. 4 (A)). The groups were formed based on a mean split. The average share of citizens served by DH across the European countries lies at 29 %. Group 1 includes on average 54 responses from countries with a below-average share of DH (questions were not mandatory, so number of ratings vary between 52 and 55). Group 2 clusters on average 33 responses from countries with an above-average share of DH.

This first group comparison shows that countries with lower shares of DH rate the importance and difficulty of social and technical barriers higher than countries with higher shares of DH. They also rate environmental barriers as more difficult to overcome. The statistically significant effects are small to medium (compare Mann-Witney-U-value, p-value and effect size r in Table 4). The highest effect (r = 0.30) is reached for overcoming technical barriers, i.e. countries with a lower share of DH

seem to face a particular challenge in tackling technical barriers to DH.

The second group comparison uses the share of renewables in DH in the countries to form the groups (cf. Fig. 4 (B)). Again, the groups were formed based on a mean split, while the mean share of renewables in DH across the European countries is 30 %. Group 1 includes countries with a below-average share of renewables in DH. It includes on average 44 responses per question. Group 2 clusters countries with an above-average share of renewables in DH and includes on average 43 responses.

The results of the second group comparison show that countries with lower shares of renewables in DH rate the importance and difficulty of economic, social and legal barriers higher than countries with higher shares. The largest effect size is found for the importance of social barriers (compare Mann-Witney-U-value, p-value and effect size r in Table 4). Here, social barriers have the highest effect (r = 0.38), i.e. countries with a lower share of renewables in DH seem to find it particularly difficult to overcome social barriers to further DH

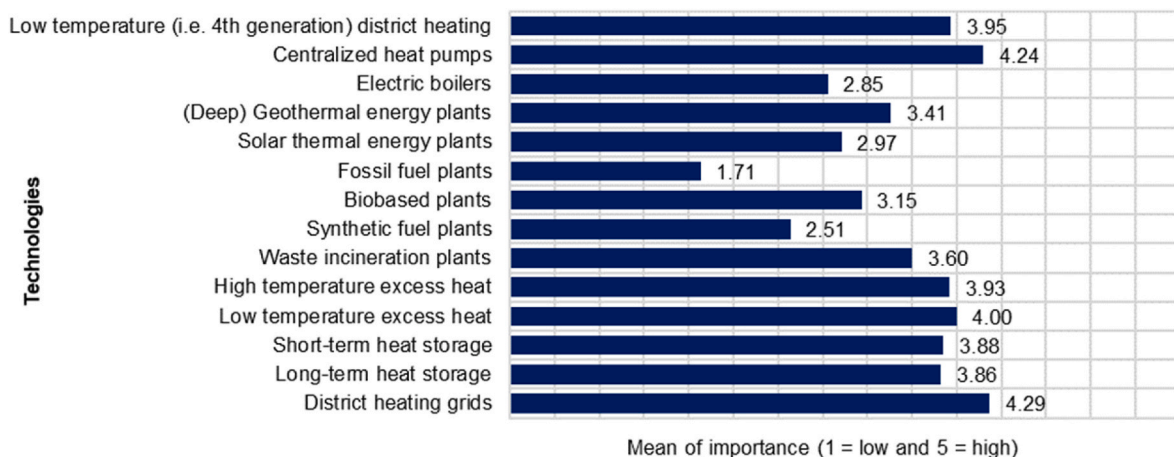


Fig. 5. Rating of importance of technologies for the further deployment of DH. Survey question: In your opinion, how important are the following technologies for the further deployment of district heating in your country? Please rate the technologies listed from 'not important' to 'very important'.

deployment.

3.2.3. Differences between technologies

Fig. 5 shows how the respondents rated the importance of different

technologies for the future deployment of DH. The related question was “In your opinion, how important are the following technologies for the further deployment of district heating in your country? Please rate the technologies listed from ‘not important’ to ‘very important’.”. District

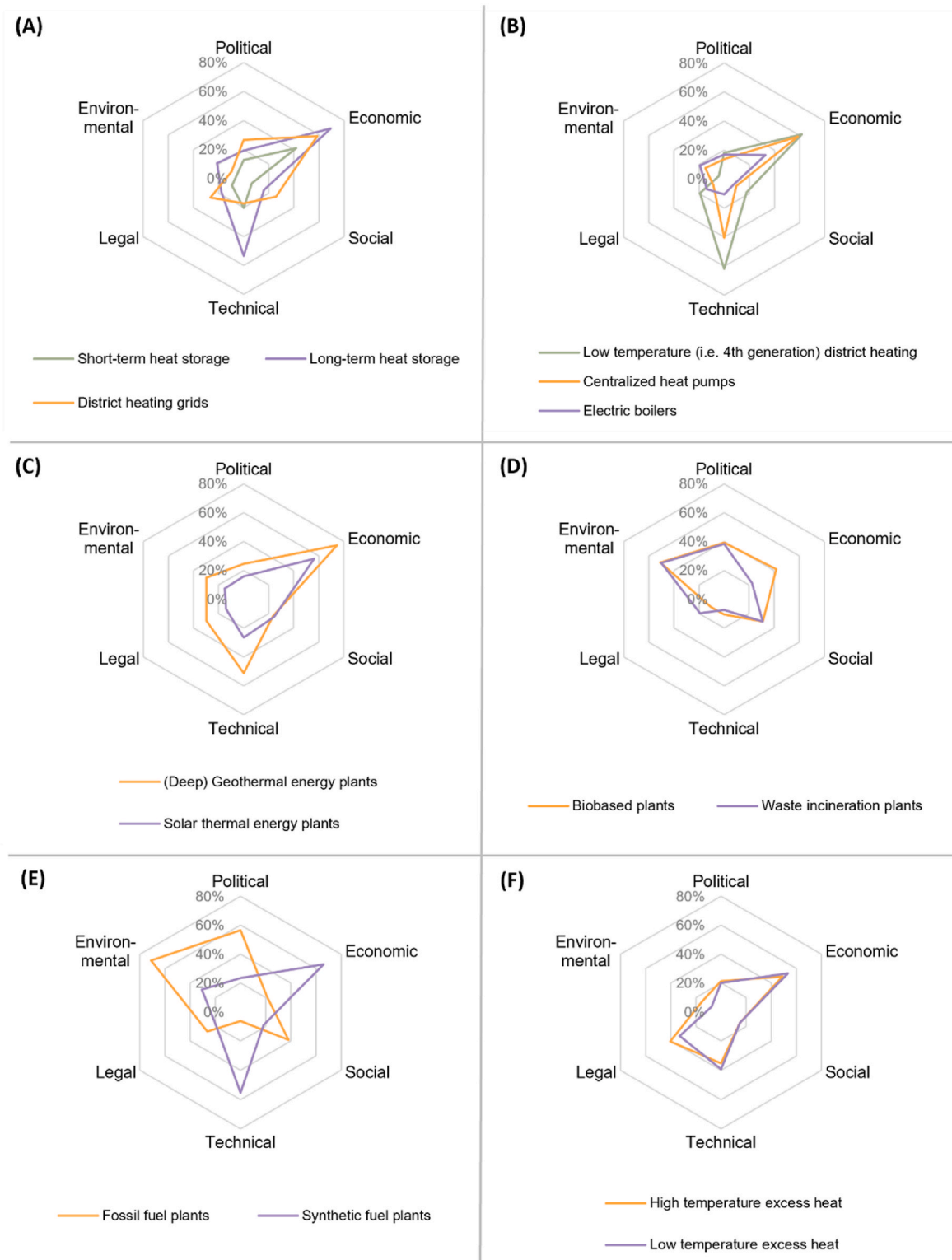


Fig. 6. Share of respondents identifying PESTLE barriers for deploying various district heating technologies. The radar plots show the percentage of respondents who reported Political, Economic, Social, Technical, Legal, and Environmental barriers for six technology groups. (A) Shows infrastructure technologies, (B) electric heat generation, (C) renewable heat generation, (D) renewable fuels, (E) fossil- and synthetic fuels and (F) the utilisation of excess heat. Survey question: Looking into barriers related to the different technologies: Which barriers are relevant to which technology in your country? Please select relevant barriers for the technologies listed.

heating grids and centralized heat pumps are perceived to be by far the most important technologies for the future of DH. On the other hand, not only fossil-fuel plants but also plants based on other fuels (synthetic fuel, solar thermal, biobased, geothermal, waste incineration and even electric boilers) are assigned a comparatively low importance. An above-average, but not critical, importance is finally assigned to excess heat and heat storage as well as fourth-generation district heating. These responses therefore imply an important shift in the importance of different fuel supply sources compared to the current status quo.

Fig. 6 shows the perceived barriers for each of these technologies according to the PESTLE framework. The question was stated as: “Looking into barriers related to the different technologies: Which barriers are relevant to which technology in your country? Please select relevant barriers for the technologies listed.”. Fig. 6 (A) reveals that economic barriers are particularly relevant for district heating infrastructure, while especially long-term heat storage also faces technical barriers. The importance of economic, and to a lesser extent technical, barriers for electric or renewable heat generation is also visible in Fig. 6 (B) and (C). Unlike the previous categories where it barely mattered, the most important barrier for bio-based or waste incineration plants is environmental as Fig. 6 (D) shows. Fig. 6 (E) illustrates that both fossil fuels and synthetic fuels face important barriers, the former environmentally and politically akin to the technologies in Fig. 6 (D) and the latter economically and technically, like other emerging technologies. Finally, Fig. 6 (F) demonstrates the importance of economic challenges to the utilisation of excess heat, with minimal differences between low-temperature and high-temperature heat.

4. Discussion of the primary district heating barriers

Based on the ranking of importance and difficulty of barriers, the highest ranked barriers are synthesised into the following key questions for district heating planners to address the deployment of DH. As found in 3.2.2, the impact of barriers varies, and some may be more relevant than others in given country or regional contexts. Despite this contextual dependency, we discuss the highest ranked barriers (cf. section 3, see Appendix A.3 for detailed rankings) in the expert survey and how these resonate with existing research at an overarching level. We end each paragraph with a brief overview of potential solutions.

(1) Is there sufficient space for the DH infrastructure?

The survey shows “limited land availability” as an important and difficult environmental barrier to overcome. DH is inherently an urban technology, facing the space and cost constraints of the urban environment (Braungardt et al., 2023). Space constraints have been identified as a barrier both underground (congestion of pipes and services) regarding DH networks (Abimbola, 2022), and above ground regarding thermal storages (Sneum, 2020; Werling, 2018).

Potential solution: A key solution is strategic local planning, allocating space for present and future needs (Braungardt et al., 2023).

(2) Is there sufficient heat demand - also long-term?

The natural monopoly characteristics of DH implies economies of scale from maximising the demand base to match the infrastructure capacity. Consequently, “low connection rates to DH” was rated as one of the most important and difficult technical barriers in the survey. Stöckel and Sneum (2023) have demonstrated this inverse relation between increased number of customers and decreasing costs per end-user. Deploying DH requires a solid business case, which again relies on solid demand data, to ensure predictability in revenues. These data (e.g. age and quality of building stock) are not always readily available (Braungardt et al., 2023), also in terms of buildings’ readiness for low temperature heat supply (Fransson et al., 2021). Millar et al. (2019) note a lack of thermal demand management, including load

prediction measures. Long-term demand uncertainty may challenge the long payback times of DH and has been raised as an issue in the UK (BREUniversity of EdinburghCentre for Sustainable Energy, 2013) and Sweden (Lygnerud, 2018). The latter study predicts that on the longer term, decreased heat demand deriving from energy efficiency will challenge the DH demand base in Europe.

Potential solution: While data- and IT-related measures were notably absent in the survey and literature, they deserve attention, as they may address the shortcomings of heat demand data and management. Longer term, DH systems may adapt to changing heat demands by diversifying into e.g. cooling and better use of resources through increased use of thermal storage (Lygnerud, 2018).

(3) Can stakeholders with legacy heating technologies coordinate effectively and adopt DH?

Path dependence can be considered both a matter arising from challenging the incumbent heat supply, and of changing the existing thinking on heat supply. The incumbency issue was raised by experts surveyed in 2022 (de la Vega and Mišech, 2022) regarding gas boilers as a challenge to renewable heating and cooling. Examples of path dependence include policymakers’ preferences for conventional DH solutions in Sweden (biofuel and waste combustion), compared to low-temperature DH (Fransson et al., 2021) and gas-based CHP in UK (Millar et al., 2019). Lowes and Woodman (2020) have identified a wider explanation for path dependence, as UK “[...] policy makers perceive heat decarbonisation as disruptive, technological pathways are seen as deeply uncertain and heat decarbonisation appears to offer policy makers little ‘up-side’”.

A related social and organisational aspect is the challenge with multiple stakeholders (BREUniversity of EdinburghCentre for Sustainable Energy, 2013). Our survey identifies the “multi-stakeholder challenge and high variety of stakeholders” as the most relevant social barrier. This describes the process of coordinating the whole ecosystem of stakeholders across space (local and national), time (development, deployment and operation phases) and sectors (e.g. coupling to electricity system). These include heat suppliers, distribution networks, incumbent utilities, end-users and the decisionmakers (policymakers and regulators). The UK Department of Energy and Climate Change (2013) additionally found that even with existing DH systems within the same city, different network ownerships introduced challenges.

Potential solution: Addressing incumbency and the deployment of DH, Devenish and Lockwood (2024) propose a comprehensive approach, coordinating “the timing and interaction of policy instruments such as heat network planning, market incentives for individual home technologies and energy efficiency interventions from the outset of the transition, and then as it unfolds.”.

(4) Is regulation stable and straightforward?

“Regulatory uncertainty” was seen as the most important and difficult legal barrier. de la Vega and Mišech (2022) similarly found experts flagging regulatory uncertainty as a challenge for renewable heating and cooling. An example of need for increased regulatory certainty, by alignment of regulation, is raised by Fransson et al. (2021) regarding the Swedish building codes and their impact on DH. Another aspect of regulatory risk was identified by BREUniversity of EdinburghCentre for Sustainable Energy (2013). Here, the UK absence of regulation for DH was noted as problematic. Developers found it difficult to reassure customers of sufficient security of supply and pricing, and thus saw merit in government-provided regulation. In terms of complexity, two large European reviews of DH framework conditions identified complex regulation as a result of diverging regulations on a sub-national level (Braungardt et al., 2023) and “Complex and partly unclear regulation for licensing” (Kocher et al., 2021).

Potential solutions: Regulating prices and setting clear standards

(Lygnerud et al., 2024), non-coercive measures to streamline permitting and improve zoning (Auvinen et al., 2023).

(5) Can stakeholders afford the investment?

The economic barriers “high upfront capital expenditure” and “high installation costs” were rated as the most important among all individual barriers. This is reflected in the large number (12) of references raising this issue, and echoed in a 2022 expert survey, finding installation costs problematic (de la Vega and Mišech, 2022). Danish cases from 2023 (Cilleborg, 2023; Jørgensen, 2023) show cancelled and more expensive projects deriving from higher-than-usual installation costs. Up-front costs have been raised as an issue in many cases, including transitioning from steam-to water-based systems (Sneum, 2020), heat pumps and geothermal (Thomaßen et al., 2021), DH expansion (Braungardt et al., 2023) and transmission and distribution (Andrews et al., 2012). Part of the up-front costs may be covered by end-users as equity in the form of connection charges. Still, too high connection fees may discourage customers (Davies and Woods, 2009), and striking the right end-user payment is important, as “... a high up-front payment might even reinforce the distrust if people feel the DH company is trying to transfer all the risk to them.” (Stöckhel and Sneum, 2023).

Potential solutions: Ranging from technical improvements to decrease upfront costs (e.g. increased modularity and scalability of geothermal (Kassem and Moscariello, 2024)) to industrial policy developing domestic components, skills and services for the DH supply chains (IRENA and REN21, 2020).

(6) Are there policies in place to integrate across sectors and overcome existing barriers (i.e. address 1–5)?

Respondents rated the “lack of system integration policies” as the most important policy barrier, followed by “missing long-term decision-making”. Whereas these two may initially look very different, what ties them together is DHs inherent ability to integrate across fuel-, electricity- and heat sectors – especially under clear and long-term policy frameworks. This link between system integration and long-term planning and policy is emphasised in the 2024 Draghi report: “Given the interaction between power and other energy vectors (such as natural gas, hydrogen, heat and carbon), network developments need to be considered in an integrated manner. A planning exercise could be developed at the EU level on grid and flexibility needs foreseeing what needs to be built in the next 20 years, building on ENTSO-E ten-year plan”. A similar importance of political barriers is indicated by Köhler et al. (2024), who reviewed barriers in literature and found the largest number of barriers within “Political framework/guiding principles”, whereas system integration was notably absent. System integration challenges have earlier been identified regarding CHP (Department of Energy and Climate Change, 2013) and power-to-heat (Sneum, 2021). Salling’s (2024) study of DH utility engineers exemplifies the link between long-term policy and system integration. She identifies a “collision between energy conservation in engineering practice and economic competition in climate policy”. In this case, it is thus not just a matter of long-term policy (here, economic competition among heat sources), but also a matter of having a policy that includes in the wider system benefits of DH (in Salling’s study the integration of excess heat from data centers). The implications of an absent long-term policy environment for DH in general is exemplified by Bolton and Foxon (2015), through the changing funding landscape, “[...] thus undermining the long term certainty required for infrastructure investments.”, and explicitly in University of Birmingham’s (2022) study of UK heat decarbonisation:

“[...] by tackling the problem piecemeal it has created a patchwork that is now too complicated, too centralised and underfunded.”.

Potential solutions: While we have referred to bespoke solutions for each sub-section above, comprehensive solutions addressing the wider set of challenges are discussed here. The European Energy Efficiency Directive² recently introduced the requirement on authorities to prepare heating and cooling plans in municipalities, establishes a framework for long-term policies that may help address points 1–4 above. The EU Heating and Cooling Strategy scheduled for 2026 (European Commission, 2025) should address these issues as well. We recommend that this and similar strategies, along with the energy planning, regulation and policies in general, should be founded not on notions like “electrify everything” or “efficiency first”, but on fundamental needs-based principles such as the energy trilemma’s equity, security and sustainability.

(7) Enough time and capital for DH deployment?

It has taken more than a century to reach 9 % global market share of DH and 13 % in Europe (IEA, 2024c; Mathiesen et al., 2023), corresponding to less than a percentage-point growth per decade. At this pace, Europe will reach its potential 2050 market shares of DH well into the 2100’s. Reaching European 2050 DH potentials would need to see growth increasing more than tenfold to 1–2 percentage-points per year, if targeting 25 %–48 % market shares (Billerbeck et al., 2024b; Mathiesen et al., 2023). A similar change of pace would likely be needed on the global heat market. All things equal, such increased deployment levels will necessitate fundamentally different approaches in energy policy. A linchpin in this fast-paced transition is its cost, as emphasised in the survey. 2050 deployment estimates for DH networks and generation in the European example reach up to 685 billion EUR (at 48 % market share) (Mathiesen et al., 2023). These numbers are substantial and raise the question where the money should come from. Popovic et al. (2024) point out that municipalities may not be able to raise such amounts of money and that DH investors – like other investors – seek the right risk-return-relationship and are cautious regarding the long investment horizons of DH. Further, areas where DH would be societally beneficial may lack deployment due to the combination of relatively low investment returns (exemplified by Gullev and Duedahl (2019)), and the risks associated with it. This gap between potentials and actual deployment is notably similar to Sorrell et al. (2006) market barriers, risk and access to capital, partially explaining the energy efficiency gap. In other words, deployment of DH hinges on the cost of capital, and the cost of capital hinges on funders’ risk perception. Four typical approaches to address this risk exist, which can be separated into economic and non-economic options. Economic incentives can be distinguished into subsidies for the desired technology or taxes on the non-desired technology. Non-economic options are mandates for the desired technologies and bans against the unwanted technology. While a traditional economic approach could be to incentivise DH deployment by subsidies or taxing the alternative, a Danish study indicated that traditional economic measures alone may not be sufficient: 57 % of gas boiler owners were unwilling to switch to alternative heat sources, even in the case of savings of 1600 EUR/year (Christiansen, 2020). In 2024, a Danish standard house with gas would pay 3000 EUR/year (Patursson and Gregersen, 2024), i.e. customers would reject savings of approximately half of their heating bill to remain with gas.

Potential solution: The presence of risk averse investors and the limited effectiveness of economic incentives for DH deployment indicate that non-economic measures are required. For fast deployment of DH, measures like zoning or mandates can ensure the planning certainty required for high-cost investments over long-term horizons in line with

² Directive (EU) 2023/1791 of the European Parliament and of the Council of 13 September 2023 on energy efficiency and amending Regulation (EU) 2023/955 (2023b).

the surveyed experts' emphasis.

5. Conclusions and policy implications

With heating and cooling accounting for half of global final energy demand and district heating (DH) recognised as a vital component of both current and future climate-neutral energy systems, this paper identifies key barriers to DH deployment. The current rate of DH deployment needs to increase substantially in order to reach European targets by 2050, but several barriers prevent such progress. We approach these barriers by studying the scientific and grey literature, and by surveying 94 experts from 20 European countries and North America, corresponding to approximately 20 % of global DH supply and 93 % of European DH supply (cf. section 2). The literature review identifies barriers, while the survey highlights the importance and difficulty of these barriers. While our paper solely focuses on barriers, not solutions, we employ the conclusion to provide a set of policy implications that may aid in getting DH "on track" in IEA terms and to expand on the Draghi report's regulation and planning-measures to accelerate deployment of DH.

The potential for successful deployment of DH depends on several critical factors. Primarily, high investment and the lack of a long-term policy framework stand out as barriers, both in the literature review and the expert survey (cf. section 3 and 4). These issues are interrelated, as the large upfront costs and long payback periods of DH infrastructure investments require long-term certainty to mitigate investor risk. Attracting investments by reducing risks is thus key, for example through zoning or technology mandates.

Furthermore, survey results confirm a lack of system integration policies (cf. section 3), indicating that the potential for sector coupling is not adequately reflected in the current policy landscape. This deficiency reduces market liquidity and system flexibility. It may be addressed by regulation valuing the wider societal benefits from DH and alternative heat sources. Specifically, energy system integration and other values brought by the different alternatives in the heat sector should be analysed (e.g. by energy system modelling (Billerbeck et al., 2024b; Sandvall et al., 2017; Spirito et al., 2024)), which in turn enables a better understanding of societally desirable heat solutions, informing long-term policymaking and regulation in line with societal priorities, and enabling a level playing-field among heat sources. This exercise can be coordinated by an integrated Energy System Operator, as suggested by Robinson and Tennican (2024). Such an operator has been implemented in UK since 2024, tasked with planning and operation across multiple energy sectors. This energy system planning across sectors departs from sector-specific planning and policy towards a whole-of-system approach.

Finally, the barrier analysis reveals the need to ensure sufficient space for implementation, and to assess long-term heat demand (cf. section 3 and 4). A key policy implication is therefore the need for transparent data to reduce uncertainties, in particular future heat demand data, which can be supported by smart metering and digitalisation. Heat planning, i.e. long-term strategies at local level to map current and future heat demand as well as resources like geothermal, excess heat and sources for large scale heat pumps, can be a useful tool to navigate potentials and uncertainties in the deployment of DH. The recent introduction of the heat planning requirement through the European Energy Efficiency Directive is a pertinent example of support for a stronger policy framework (cf. section 4). For such planning to be in line with the surveyed experts' view on the most relevant future DH technologies, beyond the deployment of DH grids themselves, it should include a strong focus on centralised large-scale heat pumps.

Our survey reveals different perceptions of barriers across countries (cf. section 3.2), highlighting the need for context-specific strategies and policies to improve DH adoption. For example, environmental, social and technical barriers require more attention in countries with lower DH adoption rates than in those with higher rates. Therefore, it is critical

that policymakers consider the current DH landscape in each country, and on the sub-national level (e.g. states, municipalities, cantons), when designing targeted regulations and policy settings.

We have undertaken a comprehensive mapping of barriers using a systematic approach, which we believe is a first in the scientific literature. However, further research is needed to deepen our understanding of these barriers and their impact on DH uptake. This paper serves as a first step, while more specific studies can provide valuable insights for DH implementation, addressing additional barriers, e.g. those related to digitalisation, while also exploring potential solutions to promote a climate-neutral and sustainable energy future. Finally, the majority of barriers were techno-economic and five of the 46 barrier categories identified in the manual review were solely found in grey literature (for the GPT-identified categories it was all). To address these two issues, peer reviewed studies can further strengthen the knowledge base by confirming the grey literature-barriers and by exploring barriers outside the techno-economic sphere. As barriers constantly develop, the authors welcome additions to the ever-expanding list we have started with this study.

CRedit authorship contribution statement

Daniel Møller Sneum: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Anna Billerbeck:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Febin Kachirayil:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Russell McKenna:** Writing – review & editing, Validation, Conceptualization.

Declaration of AI and AI-assisted technologies

During the preparation of this work, the authors used "deepl write" to correct potential grammar mistakes and improve readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. Further, and as explained in the main body of the text, the GPT4All software was applied to review literature. The GPT generated answers are provided in the appendix, while the main article is written by the authors themselves.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendices.

A.1 Survey questions

Table A1
Overview of survey questions

Question	Answer options
1. Please select your country.	Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, United Kingdom, Iceland, Norway, Switzerland, United States
2. What is your core activity? Please select an answer.	District heating utility Non-district heating utility (e.g. within gas or electricity) Academia and research (e.g. university and independent research institute) Industry (e.g. hardware manufacturer) Consulting firm Non-governmental organization (e.g. think tank) Public authority and government Others: [free text] Free text
3. In your opinion, what are the main barriers to the further deployment of district heating in your country?	
4. How significant/important do you think the following barriers are for the further deployment of district heating in your country? Please rate the barriers listed from 'not important' to 'very important'.	Political barriers (e.g. not a focus of political strategies, no state-based financial support available) Economic barriers (e.g. high investment and operational costs, high end-consumer prices) Social barriers (e.g. low acceptance and awareness of consumers, high variety of stakeholders, skills gap)
5. How difficult do you think it is to address or overcome the following barriers for the further deployment of district heating in your country? Please rate the barriers listed from 'not difficult' to 'very difficult'.	Technical barriers (e.g. low technical maturity, low suitability of the building stock) Legal barriers (e.g. difficult permitting process, hindering market legislation or building regulations) Environmental barriers (e.g. no available resources, high air pollutant emissions) Low temperature (i.e. 4th generation) district heating Centralized heat pumps Electric boilers Geothermal energy plants Solar thermal energy plants Fossil fuel plants Biobased plants Synthetic fuel plants (e.g. hydrogen) Waste incineration plants High temperature excess heat (e.g. from industry) Low temperature excess heat (e.g. data centers, supermarkets, metro, waste water) Short-term heat storage Long-term heat storage (i.e. seasonal) District heating grids (e.g. pipes) Barriers and technologies from above
6. In your opinion, how important are the following technologies for the further deployment of district heating in your country? Please rate the technologies listed from 'not important' to 'very important'.	
7. Looking into barriers related to the different technologies: Which barriers are relevant to which technology in your country? Please select relevant barriers for the technologies listed.	
8. Would you be interested in a deep dive into the barriers and answering even more questions (it will take you about 5 more minutes)?	Yes, I would like to answer a few more questions. No, I would like to finish the survey now.
9. Looking into barriers in more detail: How significant/important do you think the following barriers are for the deployment of district heating in your country? Please rate the barriers listed from 'not important' to 'very important'.	Political barriers Not a focus of political strategies, i.e. missing prioritization Missing long-term decision making Lacking system integration policies Limited financial support is available Complicated financial support schemes
10. Looking into barriers in more detail: How difficult do you think it is to address or overcome the following barriers for the future deployment of district heating in your country? Please rate the barriers listed from 'not difficult' to 'very difficult'.	Economic barriers High upfront capital expenditure High finance costs High installation costs (e.g. labor, certification) High operational costs High end-consumer prices High costs for backup capacity Lacking financial experience (e.g. of operators, planners) Social barriers Low acceptance of consumers Low awareness of consumers of benefits Status quo bias

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Table A1 (continued)

Question	Answer options
	Multi-stakeholder challenge and high variety of stakeholders
	Missing knowledge and skills gap (of operators, installers, planners)
	Technical barriers
	Low technical maturity
	Low market maturity
	Competition between supply technologies
	Lock-in effect and path dependency
	Low suitability of the building stock
	Low connection rates to DH grids
	Legal barriers
	Regulatory uncertainty
	Difficult permitting process and administrative burden
	Lacking standardization
	Lacking capacity of authorities
	Bans or mandates for district heating
	No third-party regulations
	Environmental barriers
	Limited resources
	Limited land availability
	High air pollutant emissions
	Limited access to the power grid
	Limited fuel availability (i.e. no biomass, H2)

A.2 Survey sample

Table A2

Overview of sample

Country	Number of respondents	Activity of respondents							
		District heating utility	Non-district heating utility	Academia and research	Industry	Consulting firm	Non-governmental organization	Public authority and government	Other
Austria	3	0	0	3	0	0	0	0	0
Belgium	0	0	0	0	0	0	0	0	0
Bulgaria	0	0	0	0	0	0	0	0	0
Croatia	1	0	0	0	0	1	0	0	0
Cyprus	0	0	0	0	0	0	0	0	0
Czech Republic	2	0	0	1	0	1	0	0	0
Denmark	14	4	0	1	1	2	1	4	1
Estonia	1	0	0	0	0	0	0	1	0
Finland	7	1	0	1	0	1	1	3	0
France	2	0	0	0	0	0	0	1	1
Germany	20	0	0	14	0	2	1	2	1
Greece	0	0	0	0	0	0	0	0	0
Hungary	3	1	0	1	0	0	1	0	0
Iceland	0	0	0	0	0	0	0	0	0
Ireland	3	1	0	1	0	1	0	0	0
Italy	2	2	0	0	0	0	0	0	0
Latvia	0	0	0	0	0	0	0	0	0
Lithuania	1	0	0	0	0	0	0	0	1
Luxembourg	1	0	0	1	0	0	0	0	0
Malta	0	0	0	0	0	0	0	0	0
Netherlands	5	0	0	1	0	0	0	3	1
Norway	2	0	0	0	0	0	1	1	0
Other	4	0	0	1	0	1	2	0	0
Poland	1	0	0	0	1	0	0	0	0
Portugal	0	0	0	0	0	0	0	0	0
Romania	0	0	0	0	0	0	0	0	0
Slovakia	2	1	0	0	0	1	0	0	0
Slovenia	0	0	0	0	0	0	0	0	0
Spain	0	0	0	0	0	0	0	0	0
Sweden	5	0	0	2	0	1	0	2	0
Switzerland	10	3	2	2	1	1	0	0	1
United Kingdom	3	1	0	1	1	0	0	0	0
United States	2	0	0	0	0	0	1	0	1
Sum	94	14	2	30	4	12	8	17	7

A.3 Ratings of barriers

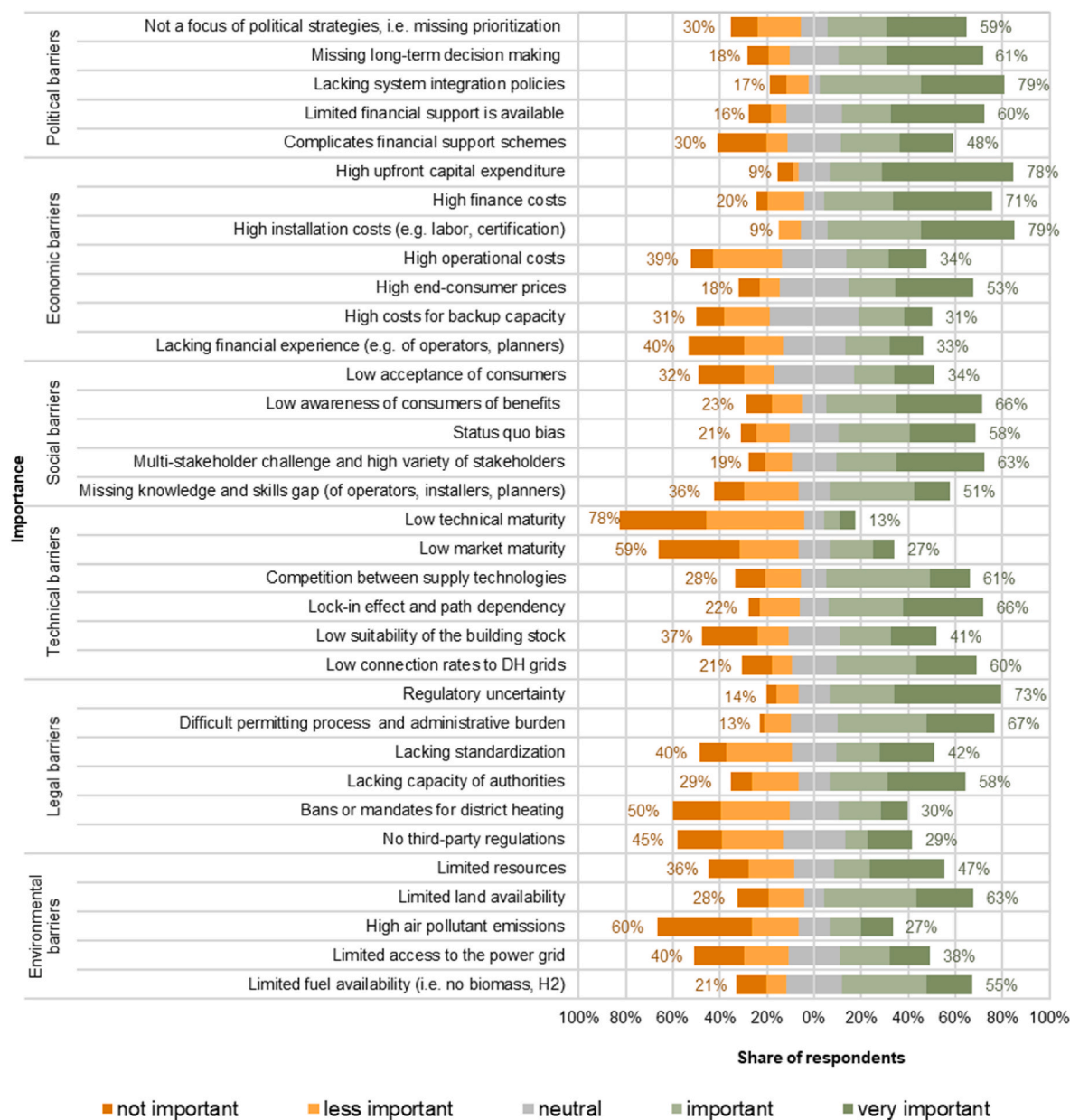


Fig. A1. Overview of rating of barriers regarding importance.

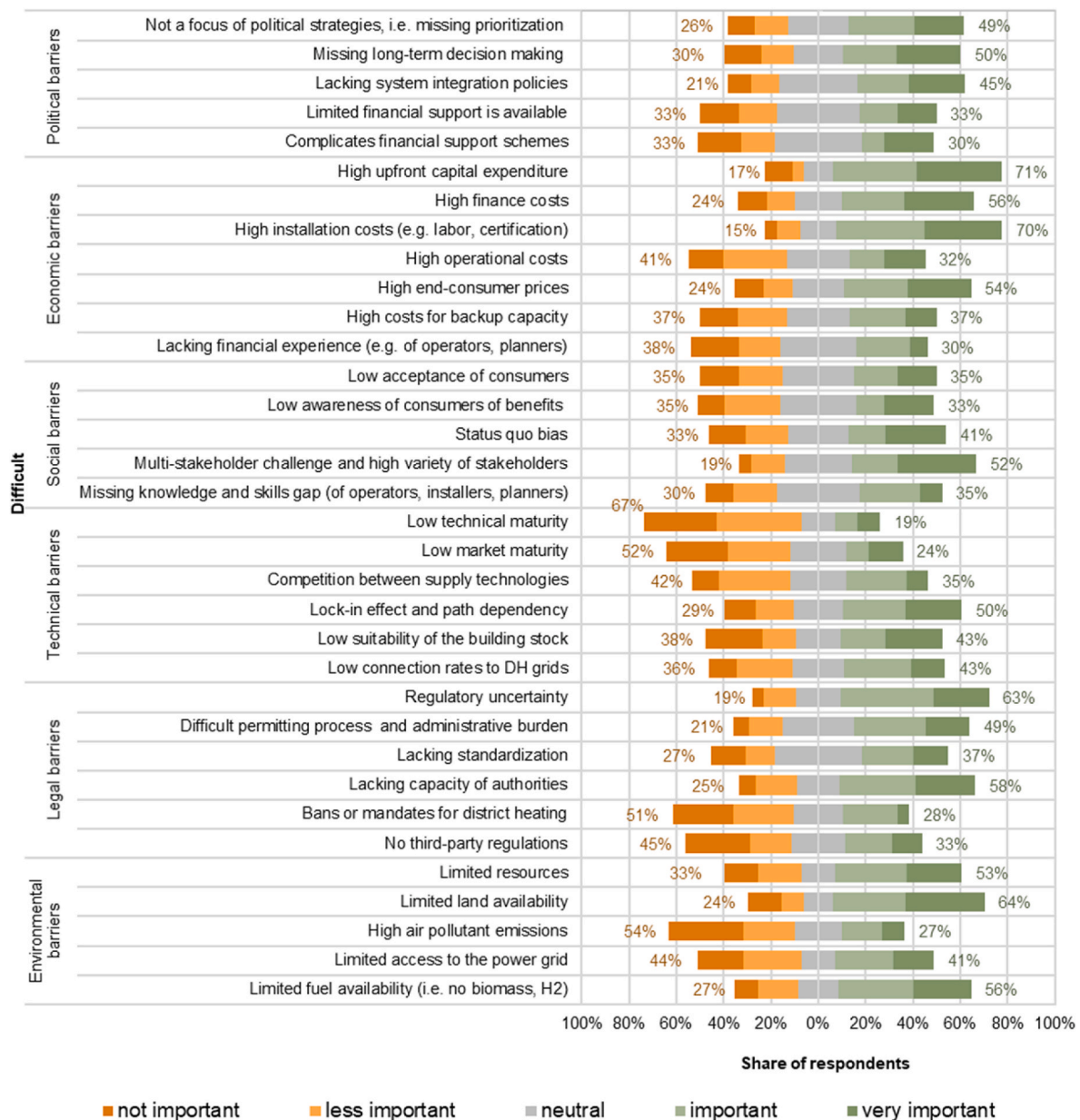


Fig. A2. Overview of rating of barriers regarding difficulty.

A.4 Extended barrier table

Table A3

Extended barrier table. Sub-categories marked with an asterisk note those that were identified in a subsequent review iteration, after the survey taking place. Table contains only information gathered in the manual review, not the GPT-based.

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Political barriers	Complicated financial support schemes	Complicated financial support schemes	District heating	EU	National	Development	Grey	Chassein and Roser (2017)
	Inconsequent application of policies to support	Inconsequent application of policies to support	District heating	EU	Plant	Operation	Peer reviewed	Marszal-Pomianowska et al. (2024)
	Lacking system integration policies	Limited market access	District heating/ CHP	United Kingdom	National	Operation	Grey	Department of Energy and Climate Change (2013)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Economic barriers	Missing long-term decision making	Local planning "forgetting" DH	District heating	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Excess heat providers currently do not have the possibility to participate in dispatch	Excess heat	Denmark	Local	Operation	Peer reviewed	Frölke et al. (2022)
		Uncertain electricity market revenue	CHP	EU	Plant	Development	Grey	Andrews et al. (2012)
		Uncertain investment environment	District heating	United Kingdom	Local	Development	Peer reviewed	Bolton and Foxon (2015)
		Conflicting policies for DH and individual HP	District heating	Denmark	National	Development	Peer reviewed	Salling (2024)
		Policy development stalled	District heating	Netherlands	National	Development	Peer reviewed	Devenish and Lockwood (2024)
		Underdeveloped policies	District heating	Germany; Netherlands	National	Development	Peer reviewed	Herreras Martínez et al. (2025)
		Need for cohesive national energy policy (IT)	Geothermal	Italy	National	Development	Peer reviewed	Meirbekova et al. (2024)
		No policy incentive for new excess heat-generating sites to incorporate excess heat recovery	Excess heat	Sweden	Plant	Development	Peer reviewed	Lygnerud et al. (2022)
		Absent policy focus	District heating	EU	International	Development	Grey	Bauknecht et al. (2017)
	Not a focus of political strategies, i.e. missing prioritization	Competition from gas	District heating	United Kingdom	National	Development	Grey	Lowes et al. (2018a)
			District heating	United Kingdom	National	Development	Peer reviewed	Millar et al. (2019)
		Lacking requirement on system integration	District heating/excess heat	Denmark	National	Development	Industry	Stærke (2022)
		Perceived complexity of heating	Heating	United Kingdom	National	Development	Peer reviewed	Lowes and Woodman (2020)
		Politically unprioritized	District heating	EU	National	Development	Grey	de la Vega and Mišech (2022)
		Rivalry between policy goals (energy efficiency and energy security)	District heating	Netherlands	Decision lvl/local	LC lvl/development	Source/peer reviewed	Gürsan et al. (2024)
		Limited financial support is available	District heating	UK	Local	Development	Grey	(BRE)University of EdinburghCentre for Sustainable Energy, 2013)
		Current incentive scheme insufficient	Geothermal	Türkiye	National	Development	Peer reviewed	Korucan et al. (2024)
		High costs for backup capacity	Excess heat	EU	Plant	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		High end-consumer prices	District heating	EU	Plant	Development	Grey	Trier et al. (2018)
	High finance costs		District heating	Netherlands	Plant	Development	Peer reviewed	Herreras Martínez et al. (2022)
		High end-user connection cost	District heating	Generic	Plant	Development	Grey	Stöckel and Sneum (2023)
		High financial return requirements	District heating	Generic	Plant	Development	Industry	Gullev and Duedahl (2019)
		Lacking finance	District heating	United Kingdom	Local	Development	Peer reviewed	Bolton and Foxon (2015)
	High installation costs (e.g. labor, certification)	Crisis-induced planning uncertainty	District heating	Denmark	National	Development	Media	Ritzau (2022)
	High operational costs	Cost of upskilling staff	District heating	United Kingdom	Plant	Development	Peer reviewed	Millar et al. (2019)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
	High upfront capital expenditure	Cost of maintaining technology is high	District heating/ heat pump	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
		High business process cost	District heating	USA	Plant	Operation	Grey	Sneum (2020)
		Cost of technology	District heating/ 4GDH	USA	Plant	Development	Grey	Sneum (2020)
			District heating	EU	National	Development	Grey	de la Vega and Mišech (2022)
		Crisis-induced price increases	District heating	Denmark	National	Development	Industry	Danish District Heating Association (2023)
		High investment for RES, heat pumps (focus on decarbonisation)	District heating	EU	International	Development	Peer reviewed	Thomaßen et al. (2021)
		High technology costs	District heating	EU	International	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
								Andrews et al. (2012)
		High thermal network investment	District heating/ thermal network	EU	Plant	Development	Grey	
		Cost of geothermal technology is high	District heating/ geothermal	Generic	Plant	Development	Peer reviewed	Mazhar et al. (2018)
	Capital lacking or unavailable*	High investment for 5GDH	5GDH	Baltics	End-user	Development	Peer reviewed	Pakere et al. (2023)
		High investment	LTDH	Italy	National	Development	Peer reviewed	Guzzini et al. (2020)
			Geothermal	Belgium; Iceland; Italy	National	Development	Peer reviewed	Meirbekova et al. (2024)
		Expensive infrastructure changes	District heating	Germany; Netherlands	National	Development	Peer reviewed	Herreras Martínez et al. (2025)
		High investor costs (drilling, risks, payback period)	Geothermal	Turkiye	National	Development	Peer reviewed	Korucan et al. (2024)
		End-user lacking capital	District heating	EU	End-user	Development	Grey	Trier et al. (2018)
		Investment capital lacking	District heating	Denmark	End-user	Development	Media	Reintoft (2023)
			District heating	United Kingdom	Plant	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
			District heating	Spain	End-user	Development	Peer reviewed	Arriazu-Ramos et al. (2021)
			District heating	China	Plant	Development	Peer reviewed	Nie et al. (2023)
		Investment capital lacking - public	District heating	EU	Local	Development	Grey	Trier et al. (2018)
		No funds for planning phase	District heating	United Kingdom	Plant	Development	Peer reviewed	Millar et al. (2019)
			District heating	United Kingdom	Local	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
	Lacking financial experience (e.g. of operators, planners)	Banks with high requirements	District heating	Germany; Netherlands	National	Development	Peer reviewed	Herreras Martínez et al. (2025)
		Lacking financial experience with technology	District heating/ 5GDH	Generic	International	Development	Peer reviewed	Gjoka et al. (2023)
		Unproven technology	District heating	EU	Plant	Development	Grey	de la Vega and Mišech (2022)
	Outdated business model*	Outdated business model	District heating	Sweden	Plant	Operation	Peer reviewed	Lygnerud (2018)
			District heating	EU	Plant	Development	Grey	Trier et al. (2018)
		5GDH requires new business models	5GDH	Baltics	End-user	Development	Peer reviewed	Pakere et al. (2023)
		Modification of existing business models for LTDH required	LTDH	Italy	National	Development	Peer reviewed	Guzzini et al. (2020)
		Simple price signal methods will not suffice	Excess heat	Denmark	Local	Operation	Peer reviewed	Frölke et al. (2022)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Social barriers	Low acceptance of consumers	Difficult to agree on price for excess heat	Excess heat	Sweden	Plant	Development	Peer reviewed	Lygnerud et al. (2022)
		Lack of feasible business models for demand response	District heating	EU	Plant	Operation	Peer reviewed	Marszal-Pomianowska et al. (2024)
		Profitability lacking	4GDH	Finland	Plant	Development	Peer reviewed	Reda et al. (2021)
		Cannibalisation among DH tech	District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
		Discrepancy in price expectations	District heating/LTDH	Sweden	National	Development	Grey	Fransson et al. (2021)
		End-user scepticism towards DH	District heating	United Kingdom	Local	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
		Lacking consumer interest in connection to DH	District heating	EU	End-user	Development	Grey	European Commission. Directorate General for Energy. et al. (2022)
		Lacking end-user protection	District heating	United Kingdom	National	Development	Peer reviewed	Millar et al. (2019)
		Poor reputation of DH	District heating	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Price trumps environment	District heating	EU Germany	Local	Development	Grey	Trier et al. (2018)
	Low awareness of consumers of benefits	High cost perception	District heating	Netherlands	End-user	Development	Peer reviewed	Zaunbrecher et al. (2016)
		Social resistance to geothermal (IT)	Geothermal	Belgium; Iceland; Italy	National	Development	Peer reviewed	Herreras Martínez et al. (2022)
		Local resistance	Geothermal	Turkiye	National	Development	Peer reviewed	Meirbekova et al. (2024)
		Bounded rationality among end-users	District heating	EU	Local	Development	Grey	Korucan et al. (2024)
			District heating	Sweden, UK	End-user	Development	Peer reviewed	European Commission. Directorate General for Energy. et al. (2023)
			Geothermal	Belgium; Iceland; Italy	National	Development	Peer reviewed	Bull and Eadson (2023)
		High cost perception	District heating/PTH	USA	Plant	Development	Grey	Meirbekova et al. (2024)
		Invisible solution	District heating	Denmark	National	Development	Grey	Sneum (2020)
		Limited awareness	District heating	Germany	Local	Development	Grey	Fransson et al. (2021)
		Unknown technology	District heating	EU	End-user	Development	Peer reviewed	Zaunbrecher et al. (2016)
	Missing knowledge and skills gap (of operators, installers, planners)	Short-termism when making investment decisions	District heating	Spain	End-user	Development	Grey	Trier et al. (2018)
		Stakeholders often unaware of excess heat recovery opportunities	Excess heat	Austria; France; Germany	National	Development	Peer reviewed	Arriazu-Ramos et al. (2021)
		Skills gap	District heating	United Kingdom	National	Development	Grey	Fritz et al. (2022)
								(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
		Building owners lack time and expertise	4GDH	Finland	End-user	Development	Peer reviewed	Reda et al. (2021)
	Multi-stakeholder challenge and high variety of stakeholders	Multi-stakeholder challenge	District heating	United Kingdom	Local	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
			District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
		Focus on a limited number of stakeholders	District heating	Netherlands	Local	Development	Peer reviewed	Gürsan et al. (2024)
		Incumbency against DH	4GDH	Finland	Local	Development	Peer reviewed	Reda et al. (2021)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Technical barriers	Status quo bias	Communication between stakeholders insufficient	Excess heat	Austria; France; Germany	National	Development	Peer reviewed	Fritz et al. (2022)
		Excess heat interruption of source's primary activity	District heating/ excess heat	EU	Plant	Development	Grey	European Commission. Directorate General for Energy, et al. (2023)
		Incumbency against DH	District heating/ thermal network	EU	Plant	Development	Grey	Andrews et al. (2012)
			District heating/ CHP	Generic	National	Development	Peer reviewed	Jenkins (1980)
			District heating	United Kingdom	Local	Development	Grey	Lowes et al. (2018b)
		Lacking economic incentive for excess heat supply	District heating/ excess heat	United Kingdom	Local	Development	Grey	(BRE)University of EdinburghCentre for Sustainable Energy, 2013)
		Limited incentive for excess heat use	District heating/ excess heat	EU	Plant	Development	Grey	European Commission. Directorate General for Energy, et al. (2023)
		Split incentives and motives	District heating	EU	Local	Operation	Grey	Trier et al. (2018)
	Competition between supply technologies	Low competitiveness	District heating	EU	National	Development	Grey	de la Vega and Mišech (2022)
		Trade-off between local efficiency and DH	District heating/ demand	EU	Local	Development	Grey	European Commission. Directorate General for Energy, et al. (2023)
		Lacking statutory access to land	District heating	United Kingdom	National	Development	Grey	Department of Energy and Climate Change (2013)
		Subsidies for competing technology	Heating/Ind HP, District heating	Denmark	National	Development	Grey	Fransson et al. (2021)
		Unlevel playing-field	District heating	EU	National	Development	Grey	Trier et al. (2018)
		Competition from gas	District heating	Netherlands	Plant	Development	Peer-reviewed	Herreras Martínez et al. (2022)
		Competition from heat pumps	Heating/Ind HP, District heating	Netherlands	Plant	Development	Peer-reviewed	Devenish and Lockwood (2024)
		Disconnection from DH to switch to individual heating	District heating	Spain	End-user	Development	Peer reviewed	Arriazu-Ramos et al. (2021)
	Cybersecurity*	Power generation may be preferred over DHC	Excess heat	Italy; Austria	Plant	Development	Peer reviewed	Santin et al. (2020)
		Cybersecurity challenges arising from increased sector coupling	District heating	Generic	Plant	Operation	Peer-reviewed	Schmidt and Leitner (2021)
	Low connection rates to DH grids Data on energy sources constrained*	Low connection rates to DH grids	District heating	Denmark	Plant	Development	Grey	Støchkel and Sneum (2023)
		Excess heat resources unknown	Excess heat	EU	National	Development	Grey	European Commission. Directorate General for Energy, et al. (2023)
			Excess heat	Latvia	National	Development	Peer reviewed	Pakere et al. (2021)
		Lack of geothermal data	Geothermal	EU	Local	Development	Grey	European Commission. Directorate General for Energy, et al. (2023)
	Lock-in effect and path dependency		Geothermal	Netherlands	Local	Development	Peer reviewed	Herreras Martínez et al. (2022)
			Geothermal	Belgium	National	Development	Peer reviewed	Meirbekova et al. (2024)
		Outdated DH solutions	LTDH	Sweden	National	Development	Grey	Fransson et al. (2021)
			District heating	United Kingdom	National	Development	Peer reviewed	Millar et al. (2019)
		Path dependence: Central solutions favoured	District heating/ LTDH	Sweden	National	Development	Grey	Fransson et al. (2021)
		Path dependence of earlier DH generations	5GDH	Baltics	End-user	Development	Peer reviewed	Pakere et al. (2023)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
		Biomass lock-in	4GDH	Finland	Plant	Development	Peer reviewed	Reda et al. (2021)
		Silo thinking among stakeholders	District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
	Long deployment phase*	PTES long initial charge	District heating/thermal storage	Generic	Plant	Development	Peer reviewed	Mazhar et al. (2018)
	Low market maturity	Supply chain for technology underdeveloped	District heating	EU	National	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
			District heating	Generic	National	Development	Peer reviewed	Mazhar et al. (2018)
		Workforce limitations	District heating	Denmark	National	Development	Industry	Stærke (2022)
		Unfamiliarity with DH technology	District heating	Germany; Netherlands	National	Development	Peer reviewed	Herrerias Martínez et al. (2025)
		Shortage of excess heat recovery know-how	Excess heat	Sweden	National	Development	Peer reviewed	Lygnerud et al. (2022)
	Low suitability of the building stock	Building stock unsuitable for DH	4GDH; 5GDH	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Poor energy performance of the existing building stock (focus on decarbonisation)	4GDH	EU	End-user	Development	Grey	European Commission. Directorate General for Energy. et al. (2022)
		Poor technical design of central heating system	District heating	Spain	End-user	Development	Peer reviewed	Arriazu-Ramos et al. (2021)
		Uncertainty regarding LTDH ability to supply existing demand	LTDH	Italy	Plant	Development	Peer reviewed	Guzzini et al. (2020)
	Low technical maturity	Inefficient operation due to limited monitoring	District heating	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Coordination of many small excess heat producers	Excess heat	Denmark	Local	Operation	Peer reviewed	Frölke et al. (2022)
		Demand for further R&D, technical deficiencies	Geothermal	Turkiye	National	Development	Peer reviewed	Korucan et al. (2024)
		Silica scaling limiting geothermal heat transfer	Geothermal	Iceland	National	Development	Peer reviewed	Meirbekova et al. (2024)
		Inflexible biomass-based boilers	4GDH	Generic	Plant	Development	Peer reviewed	Jodeiri et al. (2022)
		Immature technology	4GDH	Finland	Plant	Development	Peer reviewed	Reda et al. (2021)
	DH requires new infrastructure, gas/elec does not*	DH requires new infrastructure, gas and electricity does not	District heating	EU	Plant	Development	Grey	Andrews et al. (2012)
		Limited opportunities for existing substation retrofitting	LTDH	Italy	Plant	Development	Peer reviewed	Guzzini et al. (2020)
	Demand uncertain*	Demand data lacking	District heating/demand	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Loss of heat demand	District heating	Sweden	Plant	Operation	Peer reviewed	Lygnerud (2018)
		Opaque building stock status	District heating/LTDH	Denmark	National	Development	Grey	Fransson et al. (2021)
		Uncertain heat long-term demand	District heating/demand	United Kingdom	Local	Development	Grey	(BRE)University of EdinburghCentre for Sustainable Energy, 2013)
			District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
		Cost recovery uncertain due to short heating season	District heating/demand	China	Local	Development	Peer reviewed	Nie et al. (2023)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Legal barriers	Demand management insufficient* Demand transparency insufficient* Source and sink not co-located*	Demand management insufficient	District heating/demand	United Kingdom	Plant	Operation	Peer reviewed	Millar et al. (2019)
		Individual metering absent	District heating	Spain	End-user	Development	Peer reviewed	Arriazu-Ramos et al. (2021)
		Excess heat source and sink not co-located	District heating/excess heat	Generic	Local	Development	Peer reviewed	Mazhar et al. (2018)
			District heating/excess heat	EU	Plant	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Geothermal source and sink not co-located	District heating/geothermal	Generic	Local	Operation	Peer reviewed	Mazhar et al. (2018)
		Maximum DHC feasibility distance depends on climate	Excess heat	Italy; Austria	Plant	Development	Peer reviewed	Santin et al. (2020)
		Long distance between heat sources and users	4GDH	Generic	Plant	Development	Peer reviewed	Jodeiri et al. (2022)
			Excess heat	Latvia	National	Development	Peer reviewed	Pakere et al. (2021)
			Excess heat	Latvia	National	Development	Peer reviewed	Pakere et al. (2021)
	Bans or mandates for district heating	Bans and mandates	District heating/PTH	Denmark	National	Development	Peer reviewed	Sneum (2021)
		Third-party access not allowed	District heating	EU	National	Operation	Grey	European Commission. Directorate General for Energy. et al. (2022)
		Stringent DHW temperature regulation	4GDH; thermal network	Switzerland	End-user	Operation	Peer reviewed	Toffanin et al. (2021)
			4GDH; thermal network	Switzerland	End-user	Operation	Peer reviewed	Toffanin et al. (2021)
		Contractual restrictions about supply temperature	LTDH	Italy	Plant	Development	Peer reviewed	Guzzini et al. (2020)
	Difficult permitting process and administrative burden	Competition from lower priced alternatives	District heating	EU	National	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Complex regulation	District heating	EU	National	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
			District heating	EU	National	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Lengthy and complex permitting process	Geothermal	Türkiye	National	Development	Peer reviewed	Korucan et al. (2024)
			Geothermal	Italy	National	Development	Peer reviewed	Meirbekova et al. (2024)
		Permitting difficult	4GDH	Finland	Plant	Development	Peer reviewed	Reda et al. (2021)
	Lacking capacity of authorities	Permitting challenges coupling between electricity and DH	District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
		Lacking competencies within public authorities	District heating	United Kingdom	Local	Development	Peer reviewed	Bolton and Foxon (2015)
			District heating	EU	National	Development	Grey	Chassein and Roser (2017)
			District heating	EU	National	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Public authority resources constrained	District heating	United Kingdom	National, local	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
			District heating	Netherlands	Local	Development	Peer reviewed	Herreras Martínez et al. (2022)
	Lacking standardization	Limited long-term feasibility studies	District heating	EU	National	Development	Grey	Trier et al. (2018)
		Lacking uniform terminology	District heating/5GDH	Generic	International	Development	Peer reviewed	Gjoka et al. (2023)
		Improper combustion leading to pollution	District heating/Waste to energy	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
		Lacking standardised contracts	District heating	United Kingdom	National	Development	Grey	(BREUniversity of EdinburghCentre for
			District heating	United Kingdom	National	Development	Grey	(BREUniversity of EdinburghCentre for

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
Environmental barriers	Lacking ex-post accountability of energy decisions No third-party regulations	Regulatory divergence	District heating	EU	Local	Development	Grey	Sustainable Energy, 2013)
		Standards for heat pricing lacking	District heating	United Kingdom	Plant	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Absence of standardisation adds risk to excess heat projects	Excess heat	Sweden	National	Development	Peer reviewed	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
		Ex-post evaluation of deployment impacts absent	District heating	Spain	End-user	Development	Peer reviewed	Lygnerud et al. (2022)
		PTH	Denmark	National	Development	Peer reviewed	Grey	Arriazu-Ramos et al. (2021)
		TPA regulation unclear	Excess heat; Solar thermal	EU	National	Development	Grey	Sneum (2021)
		Current regulations make it difficult for third parties to gain access to the DH network	Excess heat	Austria; France; Germany	National	Development	Peer reviewed	European Commission. Directorate General for Energy. et al. (2023)
	Regulatory uncertainty	Discontinued economy-of-scale regulation	District heating	Denmark	National	Development	Grey	Fritz et al. (2022)
		Unpredictable building codes	District heating	Sweden	National	Development	Grey	Fransson et al. (2021)
		Regulatory risk	District heating	EU	National	Development	Grey	Fransson et al. (2021)
			District heating	Netherlands	National	Development	Peer-reviewed	de la Vega and Mišech (2022)
			District heating	Generic	Local	Development	Peer reviewed	Devenish and Lockwood (2024)
			District heating	EU	Plant	Operation	Peer reviewed	Schmidt and Leitner (2021)
		Lack of regulatory frameworks supporting demand response	District heating	EU	Plant	Operation	Peer reviewed	Marszał-Pomianowska et al. (2024)
		Lacking certainty on contracts	Excess heat	Austria; France; Germany	National	Development	Peer reviewed	
	Negative environmental externalities	High air pollutant emissions	District heating	Generic	Plant	Operation	Grey	Fritz et al. (2022)
		Induced seismicity	Geothermal	Belgium	National	Development	Peer reviewed	IRENA (2017)
	Limited access to the power grid	Electricity grid constraint	District heating/heat pump	Generic	Plant	Development	Grey	Meirbekova et al. (2024)
		Limited grid access	District heating/CHP	Generic	National	Operation	Grey	Heat Pump Centre (2019)
	Limited land availability	Lack of space	District heating/thermal storage	USA	Plant	Development	Grey	Dixon and Gielen (2006)
		Limited space for pit thermal storages	District heating/PTES	Denmark	Local	Development	Grey	Sneum (2020)
		Space availability limited	District heating/solar thermal	EU	Local	Development	Grey	Werling (2018)
			District heating/geothermal	Netherlands	Local	Development	Peer reviewed	European Commission. Directorate General for Energy. et al. (2023)
	Limited resources	Excess heat source quality mixed	District heating/Excess heat	Generic	Local	Operation	Peer reviewed	Herreras Martínez et al. (2022)
		Geothermal may generate low-grade heat	District heating/geothermal	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
		Hydrogen infrastructure missing for CHP	District heating/CHP	Generic	National	Development	Peer reviewed	Mazhar et al. (2018)
		Resource availability limited	District heating	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
			District heating/Waste to energy	Generic	National	Development	Peer reviewed	Mazhar et al. (2018)

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Table A3 (continued)

PESTLE	Sub-category	Barrier name	Technology	Geography	Level of origin	Lifecycle	Source	Reference
			District heating/ heat pump, District heating/ geothermal, District heating/ biomass	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
			District heating/ heat pump, District heating/ geothermal, District heating/ geothermal	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Solar thermal may generate low-grade heat	District heating/ solar thermal	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
		Variability of renewable sources	District heating/ solar thermal	Generic	Plant	Operation	Peer reviewed	Mazhar et al. (2018)
			4GDH	Finland	Plant	Development	Peer reviewed	Reda et al. (2021)
		Lacking energy sources	District heating	EU	Local	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Uncertain supply of waste as fuel	District heating/ Waste to energy	United Kingdom	Local	Development	Grey	(BREUniversity of EdinburghCentre for Sustainable Energy, 2013)
		Unreliable TPA suppliers	District heating/ excess heat	EU	Plant	Development	Grey	European Commission. Directorate General for Energy. et al. (2023)
		Resource availability limited due to non-acceptance	District heating/ biomass-based technologies	Netherlands	Plant	Development	Peer reviewed	Herreras Martínez et al. (2022)
		Excess heat source long-term uncertainty	District heating	Generic	Local	Development	Peer reviewed	Schmidt and Leitner (2021)
	Excavation problematic*	Polluted ground excavation cost	District heating/ thermal network	Denmark	Local	Development	Grey	Viuff (2024)
	Limited fuel availability	Limited fuel availability	District heating	EU	Local	Operation	Peer reviewed	Lake et al. (2017)

A.5 Prompts and responses from GPT4All

The GPT model was prompted with each of the six questions shown in the sub-headings. The content under each heading is the exact text generated by the GPT. The GPT provided references for the literature on which it built the response. This has been included as references in each section.

(1) What are the main political barriers to district heating deployment?

The main political barriers to district heating (DH) deployment can be categorized into several areas:

1. Lack of clear policy and regulation: In many countries, there is a lack of specific policies and regulations supporting DH development, which creates uncertainty for investors and developers.
2. Insufficient funding: Governments may not provide sufficient funding or incentives to support the development of new district heating networks, making it difficult for projects to secure financing.
3. Limited public awareness and understanding: Many people are unfamiliar with the benefits and potential of DH systems, which can lead to a lack of public support and engagement in the process.
4. Competing energy sources: The dominance of other energy sources (e.g., natural gas) or competing heating technologies (e.g., individual heat pumps) may create barriers for DH adoption.
5. Zoning laws and planning regulations: Zoning laws and planning regulations can sometimes hinder the development of new district heating networks, particularly in areas with strict building codes.

To overcome these political barriers, governments and policymakers should consider implementing policies that:

1. Encourage investment: Offer tax incentives or low-interest loans to encourage private sector investment in DH projects.
2. Develop clear regulations: Establish specific laws and guidelines for the development of district heating networks, including standards for safety, efficiency, and environmental impact.
3. Raise public awareness: Launch education campaigns to inform citizens about the benefits of district heating systems, such as improved energy efficiency, reduced greenhouse gas emissions, and enhanced community resilience.
4. Promote collaboration: Foster partnerships between government agencies, private companies, and local communities to facilitate cooperation and knowledge-sharing in DH development.

Some examples of successful policies that have helped overcome political barriers include:

1. The UK's Renewable Heat Incentive (RHI) scheme, which provides financial support for households and businesses adopting renewable heat technologies.
2. Sweden's district heating policy, which has led to the widespread adoption of this technology in urban areas through a combination of government incentives and public-private partnerships.

By understanding these political barriers and implementing targeted policies, governments can create an environment that encourages the development and deployment of efficient and sustainable district heating systems.

References (Abimbola, 2022; BEIS, 2020; BREUniversity of EdinburghCentre for Sustainable Energy, 2013; IRENA and REN21, 2020):

(2) What are the main economic barriers to district heating deployment?

Based on various reports and studies, some of the main economic barriers to district heating (DH) deployment include:

1. High upfront costs: Building a DH network requires significant investment in infrastructure, including pipes, heat exchangers, and other equipment.
2. Fluctuating fuel prices: Changes in natural gas or biomass fuel prices can affect the competitiveness of DH compared to individual space heating systems.
3. Low economies of scale: Small-scale DH networks may not be able to take advantage of the same cost savings as larger networks.
4. High debt levels for households and businesses: In some countries, such as Latvia (as mentioned in one of the reports), district heating clients have accumulated significant debts due to high heat prices.
5. Lack of investment incentives: Insufficient government support or tax breaks can discourage private investors from investing in DH infrastructure.

These economic barriers can make it challenging for districts to adopt and expand their DH systems, particularly when compared to individual space heating solutions that may be perceived as more cost-effective.

References (Abimbola, 2022; BEIS, 2020; BREUniversity of EdinburghCentre for Sustainable Energy, 2013; Euroheat & Power, 2015; European Commission. Directorate General for Energy et al., 2022; Hanna et al., 2016; IRENA and REN21, 2020):

(3) What are the main social barriers to district heating deployment?

Based on various reports and studies, some of the main social barriers to district heating (DH) deployment include:

1. Public perception and acceptance: Some people may be hesitant or resistant to adopting DH due to concerns about shared heat sources, perceived lack of control over their own energy costs, or unfamiliarity with the technology.
2. Lack of awareness and education: Limited understanding among building occupants, property owners, and local authorities about the benefits and feasibility of DH can hinder its adoption.
3. Resistance from existing stakeholders: Existing heating systems, such as individual gas boilers or electric heaters, may have a strong vested interest in maintaining their market share, making it difficult for new district heating solutions to gain traction.
4. Social housing concerns: In some cases, social housing providers and residents might be concerned about the impact of DH on fuel poverty, heat prices, or other issues related to energy affordability.
5. Community engagement and buy-in: Securing community support and acceptance is crucial for successful district heating projects. However, this can sometimes be challenging due to concerns about shared costs, reliability, or perceived lack of control over one's own energy usage.

These social barriers highlight the importance of effective communication, education, and stakeholder engagement in promoting the adoption of district heating solutions.

References (BEIS, 2020; BREUniversity of EdinburghCentre for Sustainable Energy, 2013):

(4) What are the main technological barriers to district heating deployment?

Based on various studies and reports, some of the main technological barriers to district heating (DH) deployment include:

1. Ageing infrastructure: Many existing DH networks were built decades ago and require significant investment to upgrade or replace them with modern systems.
2. Low efficiency: Older DH systems often have low thermal efficiencies, which can lead to energy losses and increased costs.
3. High operating temperatures: Traditional DH systems operate at high temperatures (around 80–90 °C), making it difficult to integrate renewable heat sources like solar thermal or biomass.
4. Incompatibility with renewables: The existing infrastructure is often not designed to accommodate the lower temperature requirements of modern renewable energy technologies, such as heat pumps or solar thermal.
5. Lack of standardization: Different countries and regions have varying standards for DH systems, which can create barriers to compatibility and integration across borders.
6. Insufficient district heating pipe networks: In some areas, there may not be sufficient infrastructure (e.g., pipes) in place to support the expansion of a DH network.
7. Inadequate heat storage facilities: The lack of suitable heat storage solutions can limit the ability to store excess heat generated by renewable sources and release it when needed.

These technological barriers need to be addressed through investments in modernization, standardization, and innovation to ensure efficient and effective district heating deployment.

References (Abimbola, 2022; BEIS, 2020; BREUniversity of EdinburghCentre for Sustainable Energy, 2013; European Commission. Directorate

General for Energy et al., 2022; IRENA and REN21, 2020):

(5) *What are the main legal or regulatory barriers to district heating deployment?*

Based on various studies and reports, some of the main legal or regulatory barriers to district heating (DH) deployment include:

1. Lack of clear regulations: In many countries, there is a lack of specific laws or regulations governing DH systems, which can create uncertainty for investors and developers.
2. Inconsistent pricing policies: Different regions have varying price structures for heat, making it difficult to compare costs across different areas.
3. Insufficient third-party access (TPA) rules: TPA allows multiple companies to use the same infrastructure, but in many countries, these regulations are not well-defined or enforced.
4. Planning and licensing issues: The process of obtaining permits for heat production facilities can be complex and time-consuming.
5. Taxation policies: Tax laws may favor other forms of energy over DH, making it less competitive.
6. Social assistance programs: Some countries have social assistance programs that provide subsidies or support to certain types of heating systems, which can create barriers for DH adoption.
7. Price regulation: In some cases, prices are regulated by the government, limiting the ability of companies to adjust their pricing strategies in response to changing market conditions.

These regulatory barriers can vary depending on the country and region, but they often contribute to a lack of clarity or consistency around district heating policies, making it more challenging for investors and developers to deploy DH systems.

Here are some specific examples from various countries:

1. Germany: The Renewable Energy Act (EEG) provides incentives for renewable energy sources, including biomass-based heat production. However, the EEG does not explicitly address district heating.
2. Sweden: Sweden has a comprehensive set of regulations governing district heating, but there is still room for improvement in terms of third-party access and pricing policies.
3. Poland: Poland's Energy Law provides some provisions related to district heating, but they are not well-defined or enforced.

These examples illustrate the need for clear and consistent regulatory frameworks that support the development and deployment of district heating systems.

(6) *What are the main environmental barriers to district heating deployment? <listaend>*

The main environmental barriers to district heating (DH) deployment can be categorized into several areas:

1. Land use and habitat disruption: The construction of DH infrastructure, such as pipes and heat exchangers, may require significant land acquisition, potentially disrupting habitats and ecosystems.
2. Resource extraction and processing: Some DH systems rely on fossil fuels or other non-renewable energy sources for power generation, which can contribute to greenhouse gas emissions and resource depletion.
3. Water usage: District heating often requires water for cooling purposes, especially in combined heat and power (CHP) plants. This can strain local water resources, particularly during periods of drought.
4. Waste management: The disposal of waste heat from DH systems can be a challenge, as it may require specialized facilities or technologies to manage effectively.
5. Air pollution: Some DH systems, especially those relying on fossil fuels or biomass combustion, can generate air pollutants like particulate matter (PM), nitrogen oxides (NOx), and volatile organic compounds (VOCs).
6. Climate change mitigation: The carbon footprint of district heating varies depending on the energy source used for power generation. Fossil fuel-based DH systems contribute to greenhouse gas emissions, while renewable-based systems can help mitigate climate change.
7. Noise pollution: District heating infrastructure, such as pipes and pumps, can generate noise that may be a concern in residential areas or other sensitive environments.

To overcome these environmental barriers, district heating developers and operators are exploring innovative solutions like:

1. Renewable energy sources: Integrating solar, wind, geothermal, or biomass power into DH systems to reduce greenhouse gas emissions.
2. Energy efficiency measures: Implementing advanced insulation materials, smart grids, and other technologies to minimize heat losses and optimize system performance.
3. Waste-to-energy conversion: Utilizing waste heat from industrial processes or municipal solid waste incineration to generate electricity or provide district heating.
4. Carbon capture and storage (CCS): Implementing CCS technology in DH power plants to reduce greenhouse gas emissions associated with fossil fuel combustion.
5. Green infrastructure design: Incorporating green roofs, urban forestry, and other nature-based solutions into DH system planning to mitigate the environmental impacts of infrastructure development.

By addressing these environmental barriers through innovative technologies and sustainable practices, district heating can become a more environmentally friendly option for providing heat and power in communities worldwide.

References (Abimbola, 2022; BEIS, 2020; BRE et al., 2013; Euroheat & Power, 2015; European Commission. Directorate General for Energy et al., 2022; Hanna et al., 2016; IRENA et al., 2020).

Data availability

Data will be made available on request.

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